

Introduction to Modern Causal Inference

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Introduction

Goals and Approach

Our goal is to get people with any amount of experience in formal mathematics past undergraduate probability to be able to take a vaguely-defined scientific question, translate it into a formal statistical problem, and solve it by constructing a tool that is optimal for the job at hand.

There are many texts and courses that share the same philosophy and have substantial overlap in terms of the pedagogic approach and topics. In particular, we owe a lot to the biostatistics curriculum at UC Berkeley and to more encyclopedic texts like [*Targeted Learning*](#). Nonetheless, we think that the particular selection of topics and tone in this book will make the content accessible enough to new audiences that it's worthwhile to recombine the material in this way.

Philosophy

This book is rooted in the philosophy of modern causal inference. What sets this philosophy apart are the three following tenets:

1. The first is that for most practical purposes, **the point of statistics *is* causal inference**. Ultimately, we humans are concerned with how to make decisions under uncertainty that lead to the best outcome. These are fundamentally causal questions that ask “*what if*” we did A instead of B? The machinery of statistical inference is agnostic to causality, but that doesn't change the fact that our motivation in using it isn't. Therefore there is little point in shying away from causal claims because noncausal claims are not often of any practical utility.
2. That leads to the second, often misunderstood point, which is that **there is no such thing as a method for causal inference**. This isn't a failing of the statistics literature so much as a failing of the naive popularization of new ideas (e.g. blog posts, low-quality papers). The process of statistical inference (for point estimation) only cares about estimating a parameter of a probability distribution and quantifying the sampling distribution of that estimate. There is nothing “causal” about that, and many are surprised to learn that the algorithms used for “causal” inference are actually identical to those used to make noncausal statements. What makes an analysis causal has little if anything

to do with the estimator or algorithm used. It has everything to do with whether or not the researcher can satisfy certain non-testable assumptions about the process that generated the observed data. It's important to keep these things (statistical inference and causal identification) separated in your mind even though they must always work together.

3. The last and perhaps greatest shortcoming of traditional statistics pedagogy is that it generally does not teach you to ask the question that makes sense scientifically and *then* translate it into a statistical formalism. This is partly because, in the past, our analytic tools were quite limited and no progress could be made unless one imposed unrealistic assumptions or changed the question to fit the existing methods. Today, however, **we have powerful tools that can then translate a given statistical problem into an optimal method for estimation** in a wide variety of settings. It's not always totally automatic, but at a minimum it provides a clear way to think about what's better and what's worse. It liberates you from the route if-this-kind-of-data-then-that-method thinking that only makes for bad statistics and bad science.

Taken together, these three points give this movement a clear, unified, and increasingly popular perspective on causal inference. We have no doubt whatsoever that this is the paradigm that will come to dominate common practice in the next decades and century.

Pedagogy

The modern approach to causal inference is already well-established and discussed at length in many high-quality resources and courses. So why have yet another book going over the same material?

Well, there are many different kinds of students with different backgrounds for whom different presentation styles work better! We've chosen an approach for this particular text that focuses on the following principles:

1. **Rigor with fewer prerequisites.** Many students are turned off or intimidated by the amount of mathematical formality that is required to rigorously understand modern causal inference. While you don't actually need a lot of math to use and understand the ideas, you do need *some* math to be able to work independently in the field. Since one of our goals is for you to be able to *construct your own* efficient estimator for a never-before-seen problem, we have to think precisely and not hand-wave away rigor.

For better or worse, the math you need is built on several layers of prerequisites- real analysis, probability theory, functional analysis, and asymptotic statistics (more on that later). In an ideal world, we would all have taken these courses and be prepared for what's next, but in practice we know many people have gaps in their knowledge.

In this book we try to address that by providing a lot of foundational background information about these topics as they come up. We do our best to build up intuition for the

component pieces of things before diving into implications. We lean heavily on figures and natural english prose to explain these concepts alongside the formal mathematical notation. We emphasize intuition and practical use rather than formalism. At a minimum, we try to be very clear about what the foundations actually *are* so that if you don't understand something you at least know where to look.

2. **Core concepts.** Modern causal inference is a huge field. There are so many exciting applications and theoretical developments that it can be hard for students to find their way at first.

We've tried to keep thing as short and concise as possible in this book by focusing on core ideas and a limited number of examples. The point of this is to emphasize what is most important and to allow you to learn the fundamentals without getting distracted. That means there are many topics that aren't covered in this book (see section below), but the idea is that this book will give you the tools and mindset you need in order to continue your learning productively.

3. **A welcoming voice.** Modern causal inference is fast-moving and intense. It's easy to feel like you don't belong or aren't good enough to participate. The agnostic, impersonal tone that is best for writing a clear academic paper doesn't always create the warm, emotional connection that is necessary for us to feel secure and ready to learn.

To deal with that problem, the voice we use throughout this book is informal and decidedly nonacademic. We write in the first person and address you, the reader, directly. Figures are hand-drawn and cartoonish. We provide personal commentary alongside agnostic technical material. All of this is deliberate and meant to help you feel comfortable and guided through this world. You are not alone in learning!

If you find this book isn't for you, that's also great! Here is a list of other resources that cover some of the same ground:

- [*Semiparametric Doubly Robust Targeted Double Machine Learning: A Review* - Edward Kennedy](#)
- [*Semiparametric Theory and Missing Data* - Anastasios Tsiatis](#)
- [*Targeted Learning* - Mark van der Laan, Sherri Rose](#)
- [*Unified Methods for Censored Longitudinal Data and Causality* - Mark van der Laan, James Robins](#)
- [*Causal Inference: What If?* - Miguel Hernán, James Robins](#)

Rigor with Fewer Prerequisites

Despite our best efforts, you will need *some* background material to get the most out of this book. The “hard” prerequisites listed are non-negotiable (this book will make no sense to you at all if you don't know what a probability distribution is), but the “soft” prerequisites are

completely optional. We don't cover these topics here because there are already so many good resources to learn them and there's no need to reinvent the wheel. We provide the curated references below so you know where to go to fill in the holes in your understanding as you go along.

Hard Prerequisites You'll need to know probability theory at the undergraduate level. Random variables, expectation, conditional expectation, probability densities should be familiar to you, and ideally you have at least a vague understanding of what convergence in distribution means.

You also need undergraduate multivariable calculus and linear algebra: integrals, derivatives, vectors, etc. should all be well-worn tools at your disposal.

If you're shaky on these topics, we recommend [Khan Academy](#). We also encourage you to check out the real analysis and measure-theoretic probability resources listed below, which will provide you with a very useful conceptual understanding.

Soft Prerequisites The following topics aren't necessary to grasp the arc of the arguments we'll make, but a fully rigorous understanding isn't possible without them. As we go along, we'll provide parenthetical commentary to elucidate concepts from these subjects that you may not be familiar with. We suggest attempting to read and understand this book all the way through and taking notes on the places where you feel like you're missing something. This will help you prioritize your time if you choose to fill in the gaps. None of this material is beyond you, I promise. It just takes a little time to work through it. I learned it all from these books myself: you may be surprised to hear I don't have any formal training in mathematical statistics!

These books and videos give what I think is the most self-contained treatment of their respective topics that is in line with the didactic approach taken in this book. They tend to have a more conversational tone, provide more background and intuition to the reader, and have plenty of worked examples. As always, however, there are many alternatives that may work better for you depending on your needs. These subjects should be tackled in the order they are listed here:

Field	Topics to Focus On	Resources
1. Intro Real Analysis	sequences, limits, series, continuity, convergence (pointwise, uniform), derivative	Understanding Analysis (Ch 1-6) - Abbott

Field	Topics to Focus On	Resources
2. Measure-Theoretic Probability	definition and construction of measure, Lebesgue integral and its properties, Radon-Nikodym theorem, change-of-variables formula, modes of convergence for random variables, independence, central limit theorem, conditional expectation	Measure Theory Youtube Lectures - Landim, Measure, Integral, and Probability - Capiński & Kopp
3. Functional Analysis in $\mathcal{L}_2$	projection, bounded dual space, Reisz representation	Introductory Functional Analysis with Applications (Ch 1-3) - Kreyszig

A few tips for self study: I recommend reading skimming each chapter first and getting an idea of what the main ideas are and how they interconnect to each other and to what you've already done. Take your time to understand what the definitions of the objects are that are used in each theorem (e.g. what exactly is a sequence / random variable / linear bounded functional?). Think of some examples and draw pictures.

Then go through the chapter again and read it slowly, line by line. Reading a line of math should take you as long as it takes to read a paragraph of english because that's often how much information is conveyed therein. Now draw some pictures or think of examples attempting the illustrate the content of the theorems. After you're done, do several of the exercises that the author provides. I've found that it also helps to make a "one-page summary" or "cheatsheet" of all the material in the chapter (definitions, important theorems, and their relationships). This will help you keep all the concepts organized as you go along.

I'll also refer to [Asymptotic Statistics](#) by A.W. van der Vaart in many places throughout the guide where we need to point to technical material. Relative to the way we write things up here, that book goes over background material much faster, so you may find it more challenging to read through. However, it does contain all the necessary information and details. If you've read the material in the table above and you really want to dot your i's and cross your t's on the material in this book you can try your hand at ch. 1, 2, 6, 7, 8, and 25 of Asymptotic Statistics.

Core Concepts

We've done our best to keep this book lean and targeted on foundational concepts. As a consequence, there are many important and interesting topics that we'll say nothing or little

about. For the sake of completeness, we'll list some of them here so you know where to start looking when you're ready to build on your solid foundation.

To keep things simple and consistent, most of the examples in this book focus on inference of the average treatment effect in simple covariate-treatment-outcome observational or randomized studies. However, there are many data structures which are much more elaborate than this. For example, we do not fully discuss inference in longitudinal settings with time-varying covariates and treatments, inference when some outcomes are censored, stochastic interventions, or continuous treatments. Non-IID data (e.g. treatment spillover in a network) is another interesting topic that is not covered. Nonetheless, what is in this book will prepare you to study all of these topics and to understand them as special cases or extensions of the general approach presented here.

While the tools we describe in this book are extremely powerful, they are not *all*-powerful. Some estimands of legitimate interest are not “pathwise differentiable” and therefore cannot be analyzed with these tools. An important example of such an estimand is the conditional average treatment effect, which is the subject of extensive research. There is not yet a unified approach to these problems.

Another interesting and useful setting that we do not discuss is online (or reinforcement) learning. Many topics related to this literature (e.g. off-policy evaluation) are directly addressable with the methods in this book, but here we omit all discussion of scenarios where the treatment policy can be manipulated during data collection as more information comes in. The framework for causal inference we develop in this book does extend to such settings but only with some added complexity that is best understood after learning the basics.

Lastly, the reader should be aware that this book discusses the most popular and widely used framework for causal inference, which is that of potential outcomes in the frequentist population model. Readers may have heard of directed acyclic graphs (DAGs) or nonparametric structural equations models (np-SEM) as alternatives to potential outcomes, but these frameworks are in fact largely equivalent for the vast majority of practical problems so it is really not worth worrying about the differences until you are very deep in the weeds.

Potential outcomes are also used outside the population model in the randomization inference framework, which differs in that only the treatment assignment is assumed to be random (i.e. sampling from a larger population is not a source of variability). Much of what you learn here is also applicable to randomization inference. Potential outcomes are also commonly used in Bayesian approaches to causal inference, which we also don't discuss here.

Topics

This book aims to be a relatively self-contained treatment of the modern approach to causal inference. As we said above, the goal is for you to be able to take a vaguely-defined scientific question, translate it into a formal statistical problem, and solve it using a tool that you

optimally construct for the job at hand. Thankfully, there's a well-developed process for doing that called the [Statistical Roadmap](#) which we follow closely in the chapters of this book:

1. To begin with, we need to first understand the process of *statistical inference*. We do that by defining a set of assumptions that represent what we know is true about the real world along with some true quantity of interest. Only then do we propose some kind of method or algorithm that, given data, produces an estimate of this quantity. We should be able to show that, with enough data, our answer gets closer and closer to the truth and that we have some way to quantify the uncertainty due to random sampling. These are the topics we'll work through in chapter 1, closing out with an example that illustrates the danger of blindly applying popular methods without thinking through the question and assumptions.
2. Once we've understood the goal and process of inference, we introduce causality. Causal inference questions are just statistical inference questions but in an imaginary world where we can always observe the result of alternative "what if"s. Of course in the real world we can't see all the "what if"s. We only get to see what happened, not what *could* have happened had we made a different choice, taken a different drug, or implemented a different policy. To link the two worlds together we have to make some assumptions to guarantee that the answer to a statistical inference problem in the real world actually represents the answer to our causal question in the world of "what if"s. Chapter 2 lays out this process (*causal identification*) and gives examples. Chapter 2 also addresses why a well-defined intervention is important to formulating something as a causal question and discusses broadly applicable methods that quantify the impact of potentially erroneous identifying assumptions.
3. Once we have an identification result in hand, we are completely done with causality and can focus on solving the real-world inference problem. But for better or for worse there are thousands of inference methods, each of which work under different assumptions and target different parameters. Choosing among these is daunting and often puts the cart before the horse because we're forced to define our question such that an existing method can solve it. Chapter 3 lays the foundation to address this problem by showing that for a wide range of assumptions and questions, there is generally a way to quantify what the best possible estimator is in terms of the *efficient influence function*. The efficient influence function completely describes the behavior of the optimal estimator in large samples. This optimal estimator always converges to the right answer with more data and has the lowest legitimate uncertainty possible under the given assumptions. We work through examples of deriving efficient influence functions for the average treatment effect in standard observational and randomized experiments.
4. Finally, chapter 4 does the hard work of putting the theory from chapter 3 into practice. Here we show how we can actually build an *efficient estimator*- one which has the efficient influence function. In other words, given a statistical problem, we can construct an optimal method to solve it. We outline the three main strategies for doing this, which are called bias-correction, estimating equations, and targeted maximum likelihood. We show the different ways in which these strategies ensure that the constructed estimator

has the right behavior. All of them ultimately attain the same goal and are theoretically equivalent in large-enough samples, but the differences are relevant for how the constructed estimators perform in practice.

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Part I

Efficiency Theory

In the previous chapters we learned how to pose a good causal question and examined what kinds of causal assumptions we might have to make in order to translate the causal estimand to an observable estimand. But once an observable estimand is established, we’re done with causality for the remainder of the analysis. At this point in a study we are left with a purely statistical problem: we want to estimate some aspect of a distribution (the estimand) given some finite samples from that distribution (the data).

The fact that identification and estimation can be cleanly separated is a wonderful feature of the modern approach to causal inference that was not widely appreciated or articulated for a long time. The causal problem is that we don’t observe the right “columns” (the potential outcomes) in the “population dataset”. The statistical problem is that we don’t observe all of the “rows”. These are fundamentally different issues that require different approaches, though there is some interplay if you get into the weeds (Aronow et al. 2025). In this part of the book we’ll dedicate ourselves to the latter, purely statistical problem.

We will begin by examining how to define whether an estimator is in some sense “optimal” for a given estimand. If you’ve already done some applied causal inference, you probably already know that there are many different practical approaches for estimating causal effects (matching, regression, inverse probability weighting, etc.). How can we judge these against each other when in reality we never observe the “truth” that all of them are trying to estimate? What reason might we have to choose one over another? Once we’ve developed a characterization of optimality, we will see how to use that knowledge to construct optimal estimators for many kinds of estimands.

As a heads up, this chapter is technical, so you will see a lot of pointers to the appendices, which explain foundational concepts in mathematical statistics. As a reminder, these are targeted at readers who have taken a course in basic probability and statistics, but perhaps not real analysis or measure-theoretic probability. The appendices by no means cover all of what there is to know about their respective topics. While I strive to be as rigorous as possible, the primary goal is not rigor but instead to give the reader a sense of why and how the concepts are most often used in asymptotic statistics so that they can begin to distinguish between the main ideas and the mathematical details in the literature.

1 Asymptotic Linearity

1.1 What do we want from estimators?

Let's forget about what we know about estimators for a second and start from scratch. What is it that we *want* from an estimator in the first place? Can we set some “deal-breaker” conditions or some “nice-to-haves”?

Recall that an estimator $\hat{\Psi}$ is a function that takes a dataset (which we represent as an empirical distribution) $\mathbb{P}_n$ and returns a number. We use $\hat{\psi} = \hat{\Psi}(\mathbb{P}_n)$ to denote the *estimate*, which is a random variable. In practice, however, the term “estimator” is often used to refer to both the function $\hat{\Psi}$ and the random variable $\hat{\psi}$ interchangeably.

Any estimator $\hat{\Psi}$ can be characterized by the distribution of $\hat{\psi}$ at each sample size n from each distribution $\mathbb{P} \in \mathcal{P}$. What would we want these sampling distributions to look like relative to the true value $\psi = \Psi(\mathbb{P})$?

Why not demand that the estimate is always *exactly* equal to the estimand, i.e. $\hat{\Psi}(\mathbb{P}_n) = \Psi(\mathbb{P})$ for all distributions $\mathbb{P}$ in the model $\mathcal{P}$? That might seem completely absurd but it's worth taking a second to ask why that is really such a crazy thing to demand. It is less extreme to ask that our estimate is correct *on average*, but even this is usually not possible to do. So let's settle for asking that our estimator is *asymptotically* unbiased: the estimate is more or less right on average given a large-enough sample size. Informally: $\mathbb{E}[\hat{\psi}] \approx \psi$ for n large enough.

If I have two estimators that are both unbiased, which is better? Since the total estimation error $\mathbb{E}[(\hat{\psi} - \psi)^2]$ can be decomposed to $(\mathbb{E}[\hat{\psi}] - \psi)^2 + \mathbb{V}[\hat{\psi}]$ (by adding and subtracting $\mathbb{E}[\hat{\psi}]$ inside the square and expanding), the unbiased estimator with the smallest overall error will be the one with the smallest variance. Therefore we should desire estimators that have small (asymptotic) sampling variance, subject to being (asymptotically) unbiased.

Let's throw in a practical consideration. When we estimate a quantity we usually report confidence intervals or p-values alongside the point estimate. Those inferential statistics depend on having some estimate of the sampling variance of our estimator, which we usually construct based on a theoretical understanding of what that sampling distribution should be. Many commonly used estimators are known to be asymptotically normal, which greatly facilitates inference. Therefore another desired criterion might be that our estimator should have a theoretically-understood sampling distribution.

Therefore a sensible list of desiderata for an estimator might be:

1. asymptotically unbiased
2. small asymptotic sampling variance
3. well-understood sampling distribution

1.2 Asymptotic Linearity

In light of these criteria, we will simplify the problem of finding an optimal estimator to finding an optimal estimator among those that are *asymptotically linear*. I'll lay out what that means, argue for why it's a reasonable restriction given our criteria above, and then explain why this restriction is helpful in characterizing optimality.

Asymptotic Linearity

Definition 1.1. As before, assume that Z_i are drawn IID from some distribution $\mathbb{P}$. An *asymptotically linear* estimator $\hat{\psi}$ is one that can be written like

$$\hat{\psi} = \psi + \frac{1}{n} \sum_i \phi(Z_i) + o_{\mathbb{P}}(n^{-1/2})$$

for some mean-zero function ϕ with mean zero $\mathbb{E}[\phi(Z)] = 0$ and finite variance $\mathbb{V}[\phi(Z)] < \infty$.

Warning

If you don't really understand what $o_{\mathbb{P}}(n^{-1/2})$ means in the definition above or why it implies the sample mean term “dominates” asymptotically, you should work through all of Appendix ?? before returning here.

The function ϕ is called the *influence function* of the estimator $\hat{\psi}$. A better name for it could be the “averaging transformation” of the data for a given estimator because once the data have been transformed by ϕ , the estimator can be closely approximated by the truth plus a sample average of those transformed variables (which have mean zero). In large samples, sample averages of mean-zero variables behave like Gaussian random noise. It's somewhat remarkable that a huge number of very complicated estimators actually behave this way, as we will see shortly. By definition, every asymptotically linear estimator has an influence function that satisfies the above criteria: having one is what makes the estimator asymptotically linear.

Example 1.1 (Sample Mean). The sample mean estimator is one example of an asymptotically linear estimator. To see that, write $\hat{\psi} = \frac{1}{n} \sum Z_i$ as

$$\hat{\psi} - \psi = \frac{1}{n} \sum \underbrace{(Z_i - \psi)}_{\phi(Z_i)} + 0.$$

For this estimator, the remainder term is exactly 0, which clearly satisfies the requirement of being $o_{\mathbb{P}}(n^{-1/2})$.

The influence function for an estimator also depends on the distribution of the data. For example, in the case of the sample mean we saw that $\phi(z) = z - \psi$ which we should really write as $\phi_{\mathbb{P}}(z) = z - \Psi(\mathbb{P})$ to emphasize that this particular influence function depends on the mean of Z *under the distribution* $\mathbb{P}$. Often we write something like $\phi_{\mathbb{P}}(z)$ or even $\phi(\mathbb{P})(z)$ to mean the influence function of a given estimator *at a particular distribution* $\mathbb{P}$. But more often than not we leave the $\mathbb{P}$ part implicit to minimize notational clutter.

In fact, it's more appropriate to say that an estimator is asymptotically linear *for some set of distributions*, i.e. within some model. But usually the restriction on the set of distributions is not particularly meaningful so we just say the estimator is “asymptotically linear (under regularity conditions)” and leave it at that.

Exercise 1.1. Let's presume that Z are drawn from a distribution $\mathbb{P}$ such that $\sqrt{n}(\frac{1}{n} \sum Z_i)$ diverges as a random variable (see Section ?? for a definition of divergence of random variables). A “textbook example” of such a distribution is $\text{Pareto}(1, 1.5)$, but this happens for many “fat-tailed” distributions. You can ask your favorite large language model if you want to see a proof.

Why does this tell you that the sample mean cannot be asymptotically linear at $\mathbb{P}$?

The point of this example isn't to convince you that the sample mean isn't asymptotically linear (it is for essentially all distributions we care about), it's just to cement the idea that asymptotic linearity of an estimator technically depends on what distributions you're talking about and to better understand the definition by considering a counterexample.

It's totally possible for two different estimators to have the same influence function (at each distribution). This is because two different algorithms can turn out to behave very similarly as the sample size increases. For example, we could construct a very silly alternative estimator for the mean by adding $1/n$ to our sample mean estimate. This extra term is certainly converging to zero at better than a $\sqrt{n}$ rate so it is negligible and the influence function for our alternative estimator is the same as for the sample mean. We'll also see the idea of two different estimators sharing an influence function come up later when we discuss different strategies for building optimal estimators: the estimators we get from using each strategy may be different, but they all have the same influence function and therefore all behave similarly in large-enough samples.

At this point you may be worried about how you can take your favorite estimator and derive its influence function. It's often possible to prove an estimator is asymptotically linear, but in this book we actually won't need to do that. We're going to go the other direction: starting from the estimand, we're going to come up with a way to determine the “best” possible influence function and then *construct* an estimator around it.

1.3 Benefits of AL Estimators

Given our desiderata, asymptotically linear (AL) estimators are a very sensible class of estimators to restrict ourselves to. For one, AL estimators are guaranteed to be consistent: that is, they converge to the estimand in probability and are thus asymptotically unbiased. This is guaranteed by the fact that any influence function ϕ is mean-zero so the associated sample mean converges to 0 by the law of large numbers.

Exercise 1.2. Consider the silly estimator $\hat{\psi} = 0$ when attempting to estimate the population mean $\mathbb{E}[Z]$ in a model containing all possible finite-mean distributions of Z . Is this estimator asymptotically linear across the entire model? Why or why not?

Another related benefit of asymptotically linear estimators is that they satisfy our practical requirement of understanding the sampling distribution. By the central limit theorem, $\psi + \mathbb{P}_n \phi$ looks more and more like $\psi + \mathbf{N}(0, \mathbb{V}[\phi(Z)]/n)$ as the sample size increases. The remaining term $o_{\mathbb{P}}(n^{-1/2})$ is negligible in comparison (by definition). More often you will see this written formally as

$$\sqrt{n}(\hat{\psi} - \psi) \rightsquigarrow \mathbf{N}(0, \sigma_*^2)$$

where $\sigma_*^2 = \mathbb{V}[\phi(Z)] = \mathbb{P}\phi^2$ (since ϕ has mean zero).

We call σ_*^2 the *asymptotic variance* of our estimator. You should think of $\hat{\psi} - \psi$ as a normal distribution that is shrinking towards a spike at 0: its variance is decreasing at a rate of $n^{-1/2}$ by the central limit theorem (see Example ?? in Section ??). We therefore “stabilize” the shrinking distribution of $\hat{\psi} - \psi$ by “blowing it up” by a factor of $\sqrt{n}$. The asymptotic variance σ_*^2 is the variance of this stable normal distribution (compare rows 1 and 3 of the figure in Figure ??). It is *not* the limit of the sampling variance $\mathbb{V}[\hat{\psi}]$ as a function of n because that limit is always zero on account of the fact that the estimate is converging to the estimand.

At any finite sample, the standard error of the estimate is roughly given by $\sigma = \frac{\sigma_*}{\sqrt{n}}$. Since the asymptotic variance is given by the variance of the influence function, understanding the influence function is all we need to understand the sampling distribution.

Constructing asymptotically valid confidence intervals and p-values for any asymptotically linear estimator is therefore as simple as estimating $\sigma_*^2 = \mathbb{P}\phi^2$. If we can construct an estimate $\hat{\sigma}_*$ that is close to σ_* for large-enough n then we can estimate the standard error of our estimate with $\hat{\sigma} = \frac{\hat{\sigma}_*}{\sqrt{n}}$ we will get asymptotic 95% coverage of ψ with an interval of the form $[\hat{\psi} - 2\hat{\sigma}/\sqrt{n}, \hat{\psi} + 2\hat{\sigma}/\sqrt{n}]$ (this is a “Wald-type” confidence interval).

Exercise 1.3 (Wald Interval).

- a. Assuming that $\sqrt{n}(\hat{\psi} - \psi) \sim \mathbf{N}(0, \sigma^2)$, prove that $\hat{\mathbf{C}} = [\hat{\psi} \pm 2\sigma/\sqrt{n}]$ is a 95% confidence interval, i.e. $\mathbb{P}\{\psi \in \hat{\mathbf{C}}\} > 0.95$.
- b. Assuming asymptotic linearity, how does our actual knowledge of the sampling

- distribution of $\hat{\psi}$ compare to what we assumed in part (a)?
- c. In the real world our confidence interval is $[\hat{\psi} \pm 2\hat{\sigma}/\sqrt{n}]$. How does this compare to what we used in part (a)?
 - d. Given (b) and (c), do you think the real-world confidence interval will generally have better-than 95% coverage for all sample sizes?

Example 1.2 (Variance of the sample mean). In the case of the sample mean estimator $\hat{\psi} = \frac{1}{n} \sum Z_i$ we previously saw the influence function was $\phi(z) = z - \psi$. A plug-in estimate of $\mathbb{P}\phi^2$ is therefore given by $\frac{1}{n} \sum \hat{\phi}(Z_i)^2 = \frac{1}{n} \sum (Z_i - \hat{\phi})^2$

Example 1.3 (Variance of ordinary least squares coefficients). Let's say we're trying to estimate the regression coefficients in a linear model. Let $(Y, X) \sim \mathbb{P}$ with $X \in \mathbb{R}^d$. Let $\beta_0 = \operatorname{argmin}_{\beta} \mathbb{E}[(Y - X\beta)^2]$ be the coefficients that minimize the population mean squared error. This will be our estimand. We haven't assumed the Y vs. X relationship is truly linear, we're just estimating the best linear approximation. As an estimator we will use the ordinary least squares (OLS) estimate $\hat{\beta} = \operatorname{argmin}_{\beta} \frac{1}{n} \sum [(Y_i - X_i\beta)^2]$.

It is a well-known result that the OLS estimator is asymptotically linear under mild regularity conditions. For one proof of this see chapter 5 of Asymptotic Statistics (but it's not important at the moment). The influence function is

$$\phi(x, y) = \mathbb{E}[XX^\top]^{-1}x(y - x^\top\beta)$$

You may recognize this because it's where the “sandwich” estimator (also known as “Huber-White” or “robust” estimator) of the standard error comes from. The (asymptotic) sandwich standard error is a plug-in estimate for the quantity $\mathbb{E}[XX^\top]^{-1}\mathbb{E}[X(Y - X^\top\beta)^2X^\top]\mathbb{E}[XX^\top]^{-1}$, which is nothing other than $\mathbb{P}\phi\phi^\top$, i.e. the equivalent of $\mathbb{P}\phi^2$ when ϕ is a vector-valued function (as is the case when the estimand is a vector).

In this example the dependence on $\mathbb{P}$ is through the expectation $\mathbb{E}[XX^\top] = \mathbb{E}_{\mathbb{P}}[XX^\top]$ and the true best regression coefficients $\beta = \beta_{\mathbb{P}}$. As usual, I've left off the explicit dependence on $\mathbb{P}$ in the notation.

The interesting thing about this example is that computing the estimate involves a fairly complex algorithm: either gradient descent or a series of matrix multiplications and inversions. It's not a “simple” estimator like the sample mean. Nonetheless, in large enough samples, the estimator behaves as if it were just a sample mean of a transformation of the data.

OLS is a special case of a huge class of estimators called M-estimators. Others include sample quantiles, robust regressions, and many others. All of these are asymptotically linear under mild regularity conditions.

In our list of desired characteristics we also wrote that we wanted estimators to have as small a variance as possible. Asymptotic linearity by itself doesn't guarantee small variance, but it does give us a very easy way to compare two estimators. If we have estimators $\hat{\psi}_1$ and $\hat{\psi}_2$ which are both asymptotically linear, then all we need to know to decide which one is better (in large-enough samples) is to compare their asymptotic variances which are given by $\mathbb{P}\phi_1^2$ and $\mathbb{P}\phi_2^2$.

You may also recognize $\mathbb{P}\phi^2$ as the squared $\mathcal{L}_2(\mathbb{P})$ norm of ϕ . Therefore we are comparing $\|\phi_1\|$ to $\|\phi_2\|$. In other words, the better estimator is whichever has the “smaller” influence function! In the following sections this is the key insight we will use to construct optimal estimators.

Warning

If you are unfamiliar with $\mathcal{L}_2$ spaces or don't understand why $\mathbb{P}\phi^2$ is a norm, you should work through Section ?? and Section ?? now.

The theory of optimality we are building is only valid for asymptotically linear estimators. If there were many practically useful estimators that were *not* AL, this could be a problem because we would not be able to usefully compare them. We'll discuss this more in a later section but the short answer is that AL estimators are an extremely broad class and we're not leaving much of anything on the table.

2 Regularity

While restricting ourselves to asymptotically linear estimators is a good start, unfortunately it is not quite enough to develop a sensible theory of optimality. In this section we'll first argue for why a further restriction is not only sensible but useful. Then we will come up with and refine a definition that suits our needs.

2.1 Hodges' Estimator

Let's presume we're interested in estimating the mean $\psi = \Psi(\mathbb{P}) = \mathbb{E}_{\mathbb{P}}[Z]$ from n IID draws of a scalar random variable Z that we know is normally distributed with standard deviation $\sigma = 1$. Our statistical model is therefore $\mathcal{P} = \{\mathbf{N}(\psi, 1) : \psi \in \mathbb{R}\}$, a one-dimensional parametric model. Obviously this is a contrived example where we know a lot about Z , but the point is to show that even in this extremely favorable setting we can be led astray by mathematical arguments that put our attention in the wrong place.

We'll consider two estimators:

$$\begin{aligned}\hat{\psi} &= \frac{1}{n} \sum Z_i \\ \tilde{\psi} &= \begin{cases} \hat{\psi} & |\hat{\psi}| > n^{-1/4} \\ 0 & |\hat{\psi}| \leq n^{-1/4} \end{cases}\end{aligned}$$

The first estimator is just the usual sample mean. The second is what is called *Hodges' estimator*. When the sample is large, Hodges' estimator almost always returns the sample mean. But when the sample is small, Hodges' estimator often ignores the sample mean and returns zero.

Let's compare the asymptotic sampling distributions of the two. Since Z are normal random variables with unit variance and mean ψ , we know that the sample mean has the exact distribution $\mathbf{N}(\psi, 1/n)$. Thus it follows that

$$\sqrt{n}(\hat{\psi} - \psi) \rightsquigarrow \mathbf{N}(0, 1)$$

Warning

If you are not familiar with convergence in distribution you should work through Appendix ?? now.

Hodges' estimator, however, has the following asymptotic distribution:

$$\sqrt{n}(\hat{\psi} - \psi) \rightsquigarrow \begin{cases} \mathbf{N}(0, 1) & \psi \neq 0 \\ \mathbf{N}(0, 0) & \psi = 0 \end{cases}$$

Derivation of the asymptotic distribution of Hodges' estimator

Let's prove that $\sqrt{n}(\tilde{\psi} - \psi) \rightsquigarrow \mathbf{N}(0, 0)$ when $Z \sim \mathbf{N}(0, 1)$. This will require some understanding of what is meant formally by convergence in probability and a common probability tool called Chebyshev's inequality. It's not important to understand this proof if you're comfortable with the conclusion, I just provide it for completeness and as a refresher for those who are familiar with this kind of argument.

Since convergence in distribution and in probability are equivalent when the limit is a point mass (see Appendix ??), the statement is the same as saying that $\sqrt{n}(\tilde{\psi} - 0) \xrightarrow{\mathbb{P}} 0$. That's what we will prove.

We can write Hodges' estimator as $\tilde{\psi} = 1(|\hat{\psi}| > n^{-1/4})\hat{\psi}$. Then, for any $\epsilon > 0$,

$$\begin{aligned} \mathbb{P}\{|\sqrt{n}\tilde{\psi}| > \epsilon\} &= \mathbb{P}\{1(|\hat{\psi}| > n^{-1/4})|\hat{\psi}| > \epsilon/\sqrt{n}\} \\ &= \mathbb{P}\{|\hat{\psi}| > \epsilon\sqrt{n} \mid |\hat{\psi}| > n^{-1/4}\} \mathbb{P}\{|\hat{\psi}| > n^{-1/4}\} \\ &\leq \mathbb{P}\{|\hat{\psi}| > n^{-1/4}\} \end{aligned}$$

by definitions and total probability. We can bound the last term by applying Chebyshev's inequality to the distribution of $\hat{\psi}$ and using the probability threshold $t = n^{-1/2}$:

$$\mathbb{P}\{|\hat{\psi}| > n^{-1/4}\} = \mathbb{P}\{|\hat{\psi}| - \mathbb{E}[\hat{\psi}] > n^{-1/4}\} \stackrel{\text{Cheb.}}{\leq} \frac{\mathbb{V}[\hat{\psi}]}{n^{-1/2}} = \frac{1}{\sqrt{n}} \rightarrow 0$$

where we used the fact that $\hat{\psi} = \frac{1}{n} \sum Z \sim \mathbf{N}(0, 1/n)$ when $Z \sim \mathbf{N}(0, 1)$.

This proves the desired convergence. Perhaps surprisingly, the proof still works if we are interested in proving that $r_n(\tilde{\psi} - \psi)$ converges in probability to 0 when the mean of Z is zero for an arbitrary rate r_n . In other words, this convergence happens arbitrarily quickly as far as the asymptotics are concerned.

Nearly identical arguments are used to show that Hodges' estimator has the same asymptotic distribution as the sampling mean when the true mean is not zero. This is a fine exercise to try on your own to practice your probability theory.

Like the sample mean, Hodges' estimator is definitely asymptotically linear: the influence function is

$$\phi_{\mathbb{P}}(z) = \begin{cases} z - \mathbb{E}[Z] & \mathbb{E}[Z] \neq 0 \\ 0 & \mathbb{E}[Z] = 0 \end{cases}$$

which is certainly a $\mathbb{P}$ -mean-zero $\mathcal{L}_2(\mathbb{P})$ function for all $\mathbb{P}$ with finite mean of Z . Therefore if we were defining optimality among AL estimators, we would have to consider Hodges' estimator as a candidate.

Exercise 2.1. Judging by the asymptotic distributions, do you think the sample mean or Hodges' estimator is better?

The sample mean and Hodges' estimator are asymptotically the same across most distributions. But in any case where $\mathbb{E}[Z] = 0$, Hodges' is clearly better: its variance goes to zero even if blown up by the square root of n whereas the sample mean's variance does not. Therefore by these arguments Hodges' is the better estimator overall.

But if this is true, why do we use the sample mean instead of Hodges' estimator? And moreover why couldn't we reason the same way about a Hodges'-style estimator that has a special preference for some other value instead of zero? Wouldn't that estimator also be better than the sample mean for that special value and otherwise the same?

The reason we don't use Hodges' estimator is that the asymptotic arguments hide that this estimator actually performs much worse than the sample mean for most values of n and most possible distributions of Z . You can see this in a simulation (or work it out analytically):

The y -axis of Figure ?? shows the relative efficiency of Hodges' estimator, defined as the ratio of MSEs between Hodges' estimator and the sample mean: $\mathbb{E}[(\tilde{\psi} - \psi)^2] / \mathbb{E}[(\hat{\psi} - \psi)^2]$. If this is greater than one, then Hodges' estimator is better than the sample mean, if less than one, it is worse.

As you can see, regardless of what n is, Hodges' estimator is worse (and sometimes *much* worse) than the sample mean for the majority of the plotted values of ψ . So then why do the asymptotic arguments suggest that Hodges' estimator is actually better than the sample mean?

The problem is that the asymptotics are only concerned with the behavior of each estimator at any particular distribution of Z . In our contrived model, the distributions of Z are all normal with unit variance so a single distribution corresponds to a value of ψ . If you look at a single nonzero point on the x -axis (say, $\psi = 1/2$) and compare the relative efficiency as n increases you'll see that it does converge to 1: both estimators are asymptotically equivalent. At the point $\psi = 0$, however, the relative efficiency converges to 0 meaning that Hodges' estimator is infinitely more efficient at that point asymptotically. We can repeat our simulations but plot

``summarise()`` has grouped output by 'n'. You can override using the ``.groups`` argument.

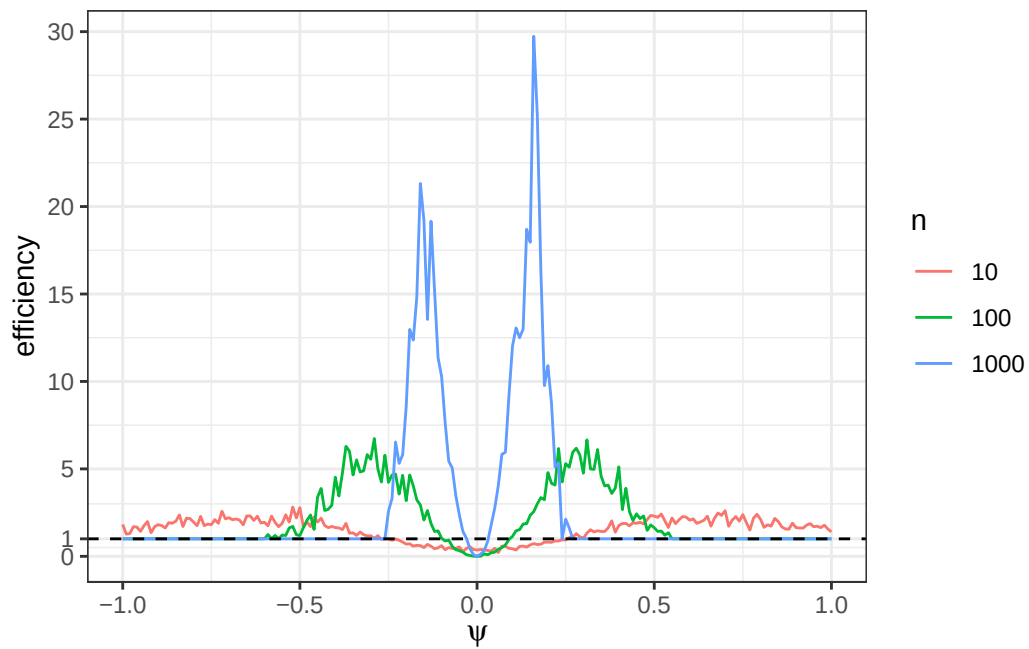


Figure 2.1: Relative efficiency of Hodges' estimator relative to the sample mean across ψ for a few values of n .

them a little differently in order to see this. Instead of plotting efficiency against ψ and slicing by n , we'll instead plot efficiency against n and slice by ψ (you could imagine a 3D plot with efficiency on the z-axis and n and ψ on the x and y).

``summarise()`` has grouped output by 'n'. You can override using the ``.groups`` argument.

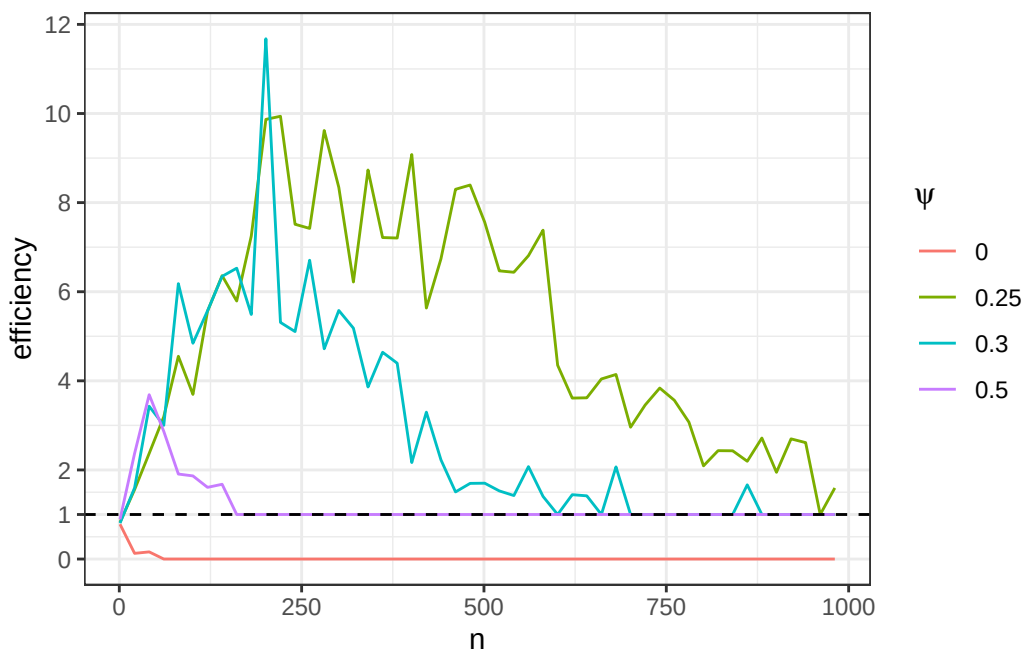


Figure 2.2: Relative efficiency of Hodges' estimator relative to the sample mean across n for a few values of ψ .

In Figure ?? you'll see that the bad behavior of Hodges' estimator is temporary for each value of $\psi \neq 0$. As n increases, the estimator eventually behaves like the sample mean. Since the asymptotics don't care about any initial behavior of the sequence, they completely ignore the fact that Hodges' estimator is usually much worse than the sample mean because those problems are "temporary" as far as the behavior at any fixed distribution is concerned.

The issue with that is that is that the problem doesn't actually disappear, it just goes to another distribution! As you can see in our first plot, the region of bad behavior simply moves as we increase n . If you're standing still at a given $\psi \neq 0$ you'd think everything is fine, but if you look across all values of ψ you notice that we're just moving the problem around. That is precisely what the asymptotic arguments ignore: they are incapable of looking at the behavior of each estimator *at multiple distributions simultaneously*. They only consider a single distribution at a time. We call this *pointwise* asymptotic behavior because we're looking at each "point" (distribution) P in the model $\mathcal{P}$ one-at-a-time. Once we look across all values of

ψ at the same time, it's clear that Hodges' estimator is much worse than the sample mean.

Asymptotic linearity is only concerned with pointwise convergence, but to prevent the bad behavior of Hodges' estimator we need some kind of *uniform* convergence: we'd like the sampling distributions of the estimates at each $\mathbb{P}$ to “arrive” or “approach” their limits at the same time. Physically you can imagine flattening a marshmallow: if you push down at a point on the marshmallow (e.g. by squeezing it between your thumb and index finger), the mass will can just go elsewhere. But if you sandwich it between your palms, you ensure that it gets squeezed down everywhere at the same time. That's “uniform” convergence.

Warning

See Section ?? for the definition of uniform convergence of functions. This is a common type of convergence that we often want to generalize for other kinds of objects. In what follows we will present a notion of “uniform” convergence in distribution for random variables.

Estimators like Hodges' estimator are called *superefficient* for reasons that will be explained in a few sections' time. In general, superefficient estimators perform better than other estimators on some very small set of distributions at the cost of arbitrarily bad performance that moves around over the other distributions as n increases. We want to rule out that sort of behavior so that we can compare estimators asymptotically without getting burned by their finite-sample behavior.

2.2 Regularity in a Parametric Model

We've identified a problem with looking only at AL estimators: there are always AL estimators that look good in terms of their pointwise asymptotics but are actually bad because they just move the problem around in a way that makes it invisible to the asymptotics.

To rule this out, we'll impose an addition condition called *regularity*. Informally, regularity means that the estimator approaches its limit distribution in a (locally) uniform way across the model so that asymptotics can't hide bad performance that moves around the model.

We will start with a toy definition of regularity that we will build up to the formal definition in the next few sections. That will require us to introduce and explain a few abstractions but these will also come in handy later so it's worth doing. For now, though, here's a first draft:

Regularity in the Normal Location Model

Definition 2.1. An estimator $\hat{\Psi}$ is *regular* at the point $\mathbb{P} = \mathcal{N}(0, 1)$ in the model $\mathcal{P} =$

$\{\mathbf{N}(\psi, 1) : \psi \in \mathbb{R}\}$ if there is a single, fixed distribution $\mathbb{D}$ such that

$$\sqrt{n} \left(\hat{\Psi}(\mathbb{P}_n^{(n)}) - \Psi(\mathbb{P}^{(n)}) \right) \rightsquigarrow \mathbb{D}$$

for any sequence of distributions $\mathbb{P}^{(n)} = \mathbf{N}(\psi_n, 1)$ such that $\sqrt{n}\psi_n$ is bounded.

First, let's clarify the notation. Since $\mathbb{P}_n$ is used to denote the (random) empirical distribution of n draws from a distribution $\mathbb{P}$, I'm using superscripts to denote a sequence of (fixed) distributions $\mathbb{P}^{(1)}, \mathbb{P}^{(2)} \dots \mathbb{P}^{(n)} \dots$. In this convention, $\mathbb{P}_n^{(m)}$ would be the empirical distribution of n draws from distribution $\mathbb{P}^{(m)}$.

Before understanding the definition above, let's break down the following standard asymptotic statement which says that $\hat{\psi}_n = \hat{\Psi}(\mathbb{P}_n)$ is a $\sqrt{n}$ -asymptotically normal estimate of $\psi = \Psi(\mathbb{P})$:

$$\sqrt{n} \left(\hat{\Psi}(\mathbb{P}_n) - \Psi(\mathbb{P}) \right) \rightsquigarrow \mathbf{N}(0, 1)$$

For each sample size n , the left-hand side here is a random variable corresponding to the scaled and centered sampling distribution of the estimator, drawing the n data points from the distribution $\mathbb{P}$. You can imagine standing still at the distribution $\mathbb{P} \in \mathcal{P}$: first drawing many datasets of size $n = 1$ and passing them each through $\hat{\Psi}$ to generate the sampling distribution of $\hat{\psi}_1 - \psi$, then drawing many datasets of size $n = 2$ to generate the distribution of $\hat{\psi}_2 - \psi$, etc. We're just (repeatedly) drawing larger and larger datasets from the same distribution to feed through the estimator and comparing the results to the estimand. The right-hand side says that those (centered and scaled) sampling distributions look more and more like the standard normal the larger n gets.

Now compare that to our definition of regularity. The only difference is that instead of $\hat{\Psi}(\mathbb{P}_n)$ we have $\hat{\Psi}(\mathbb{P}_n^{(n)})$ and instead of $\Psi(\mathbb{P})$ we have $\Psi(\mathbb{P}^{(n)})$. What that means is that we are no longer necessarily “standing still” at a single distribution $\mathbb{P} \in \mathcal{P}$ while drawing larger datasets. Instead, we are typically *moving closer* to $\mathbb{P}$ as we increase the size of our draws.

Concretely, we start at $\mathbb{P}^{(1)} = \mathbf{N}(\psi_1, 1)$. We draw many datasets of size $n = 1$ from that distribution and pass them each through $\hat{\Psi}$, comparing the results to the current ground truth ψ_1 . Now we move to $\mathbb{P}^{(2)} = \mathbf{N}(\psi_2, 1)$. We draw many datasets of size $n = 2$ from $\mathbb{P}^{(2)}$ and pass them each through $\hat{\Psi}$, comparing the results to the *new* ground truth ψ_2 . We continue like this: at each n , we arrive at the distribution $\mathbb{P}^{(n)} = \mathbf{N}(\psi_n, 1)$, draw many datasets of size n , and generate the sampling distribution of the estimation error relative to the current ground truth. An estimator is said to be regular in the normal location model at the standard normal if that error distribution, scaled by $\sqrt{n}$, approaches a fixed distribution no matter how we choose the sequence of “truths” ψ_n as long as $\sqrt{n}\psi_n$ stays bounded.

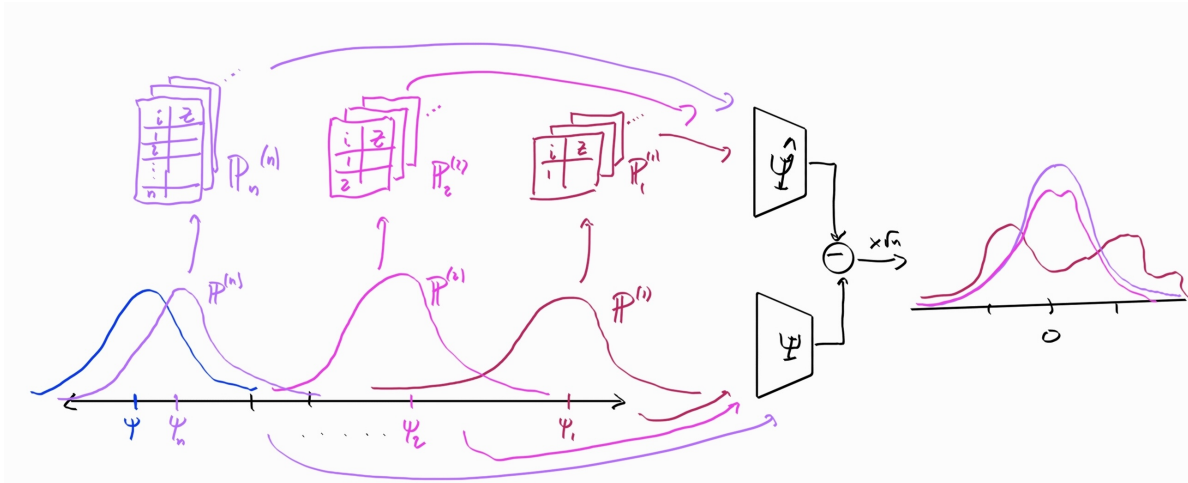


Figure 2.3: The asymptotics in the definition of regularity.

Draw a figure based on Figure ?? the depicts the standard asymptotic statement

$$\sqrt{n} \left(\hat{\Psi}(\mathbb{P}_n) - \Psi(\mathbb{P}) \right) \rightsquigarrow \mathcal{N}(0, 1).$$

What parts of Figure ?? do you have to modify to accomplish this?

To summarize: the process in the definition of regularity is the same process as in the standard $\sqrt{n}$ asymptotic convergence statement. The only difference is that we're now allowed to shift the data generating distribution under the estimator and estimand rather than leaving it fixed. In the definition of regularity, however, the right-hand side must always be the same fixed distribution $\mathbb{D}$. In other words, no matter how we move the data-generating distribution around in a shrinking neighborhood of $\mathbb{P}$ while we increase the sample size, the estimator behaves more and more like it would if we stood still at $\mathbb{P}$. The fact that the limit is the same regardless of whether we're moving the ground truth distribution around (as long as it “converges quickly” to the true distribution) is the key to regularity. Satisfying that property is what forces our estimator to have (locally) uniformly good convergence across distributions and prevents the bad behavior of Hodges’ estimator.

In our definition we only consider sequences ψ_n such that $\sqrt{n}\psi_n$ stays bounded. This means that ψ_n is converging to the reference value $\psi = 0$ relatively quickly because even if we “blow up” ψ_n by an increasing factor of $\sqrt{n}$, the result *still* converges to zero. The fact that regularity is only concerned with *quickly*-converging sequences makes it more *lenient* than if we required the same kind of asymptotic convergence in distribution along *all* sequences $\psi_n \rightarrow \psi$. We want to a-priori rule out estimators whose bad performance will escape asymptotic detection, but we want to do this in the most lenient way possible so that we don’t accidentally rule out some good estimators too. That’s why we say that regularity enforces a kind of *local* uniform

convergence in distribution: all the sequences of distributions we want to check still converge (quickly) to the distribution of interest. Another way to think of this is that the sequences we care about are all perturbations within a “ $\sqrt{n}$ -sized” neighborhood of the distribution of interest.

Exercise 2.2. Prove that the sample mean is a regular estimator at any distribution $N(\mu, 1)$ according to Definition ??:

- If $\mathbb{P}$ is such that $Z_i \stackrel{\text{IID}}{\sim} N(\mu, \sigma^2)$ for any $\mu \in \mathbb{R}, \sigma > 0$, what is the exact distribution of $\hat{\Psi}(\mathbb{P}_n) = \frac{1}{n} \sum Z_i$?
- Using your answer from part (a), what is the exact distribution of $\sqrt{n}(\hat{\Psi}(\mathbb{P}_n^{(n)}) - \Psi(\mathbb{P}^{(n)}))$ at any given n , assuming $\mathbb{P}^{(n)} = N(\psi_n, 1)$? What is the CDF $F_n(z)$ of this distribution at each n ?
- Recall that pointwise convergence of the CDFs is the definition of convergence in distribution (Definition ??). Let $\sqrt{n}(\psi_n - \psi) \rightarrow 0$ (as examples you can try $\psi_n = \psi + \frac{1}{n}$ or $\psi_n = \psi$). Prove that under any such sequence, the sequence of CDFs $F_n(z)$ from part (b) converge at each point to the same limit $F_{N(0,1)}(z)$, the CDF of the standard normal.

Exercise 2.3. In a derivation above we proved that Hodges’ estimator has excellent asymptotic properties if we stand still at $\mathbb{P}^{(n)} = N(0, 1)$:

$$\sqrt{n}(\tilde{\Psi}(\mathbb{P}_n^{(n)}) - \Psi(\mathbb{P}^{(n)})) \rightsquigarrow N(0, 0).$$

Now we’ll prove that Hodges’ estimator is *not* a regular estimator according to this definition by showing that

$$\sqrt{n}(\tilde{\Psi}(\mathbb{P}_n^{(n)}) - \Psi(\mathbb{P}^{(n)})) \rightsquigarrow N(0, 1)$$

if we move toward the true distribution like $\mathbb{P}^{(n)} = N(\frac{1}{n}, 1)$.

- Presume that $\hat{\psi} \sim N(\mu, \sigma^2)$. Given that $\tilde{\psi} = 1(|\hat{\psi}| > n^{-1/4})\hat{\psi}$, what is the CDF of Hodges’ estimate $\tilde{\psi}$ for any value of n ? It may help you to draw a picture of the “density function” for Hodges’ estimator for some fixed n (relative to the normal it has a spike at 0 comprised of the mass that was in the tails $|z| > n^{-1/4}$).
- Using your answer from part (a), what is the exact distribution of $\sqrt{n}(\tilde{\Psi}(\mathbb{P}_n^{(n)}) - \Psi(\mathbb{P}^{(n)}))$ at any given n , assuming $\mathbb{P}^{(n)} = N(\frac{1}{n}, 1)$? What is the CDF $F_n(z)$ of this distribution at each n ?
- Argue that this CDF $F_n(z)$ converges pointwise to the limit $F_{N(0,1)}(z)$, the CDF of the standard normal.

Since the asymptotic distribution of Hodges’ estimator is different depending on how we approach $N(0, 1)$ with $\sqrt{n}\psi_n \rightarrow 0$ (standing still at $\psi_n = 0$ or with $\psi_n = \frac{1}{n}$), we conclude that Hodges’ estimator is not regular at this distribution.

The definition we’re working with at the moment is specific to the normal model $\mathcal{P} = \{\mathbf{N}(\psi, 1) : \psi \in \mathbb{R}\}$. We built a very specific sequence of distributions $\mathbb{P}^{(n)} = \mathbf{N}(\psi_n, 1)$ that get closer and closer to the true distribution. Moreover we required the “distance” to the true distribution to close at the $n^{-1/2}$ rate or faster. Our challenge in the proceeding sections will be figuring out a sensible way to generalize our definition to nonparametric models. But to do that we’ll first have to establish some notions of “geometry” in that setting.

2.3 Structuring Statistical Models

Our goal at our moment is to generalize our definition of regularity by establishing some “geometry” in our model so we can talk about a sequence of distributions converging in some required sense. In our current definition, a model $\mathcal{P}$ is just a set of distributions: there is no sense of direction or distance that is inherent to it. I think of a naked model like that as a big soup of distributions: everything is free to jumble around. We need a little more structure to make sense of regularity in a nonparametric model.

For the remainder of this section we’ll work within models where every distribution $\mathbb{P}$ has a density function p . Formally, we will assume that all our distributions are absolutely continuous with respect to Lebesgue measure (don’t worry if you don’t know what that means). The results we will discuss actually do not depend on that assumption: the theory works exactly the same way for general models. However, the intuition is harder to get across in the general case, especially for readers who are new to measure-theoretic probability. I’ll abuse notation where convenient and use $\mathcal{P}$ to refer to the set of densities p as well as the set of distributions (measures) $\mathbb{P}$. If you know some measure theory, you can for now imagine that all distributions in the model have a density with respect to an arbitrary dominating measure $\bar{\mathbb{P}}$ and replace references to $\mathcal{L}_2$ below with $\mathcal{L}_2(\bar{\mathbb{P}})$. Throughout the rest of the book I will also write integrals like $\int f(z) p(z) dz$ with the standard differential dz so that they are comprehensible without measure theory. If you know measure theory, you can replace these with Lebesgue integrals with respect to the relevant dominating measure.

The first step to imposing a structure that will be useful to us is to think in terms of density functions rather than distributions (measures). If all the elements of the model have densities, we can think of the model as a collection of densities. Since densities are functions, we might be tempted to say they are elements of $\mathcal{L}_2$ because then everything is great: we have lengths, we have angles, we have all sorts of useful identities and relationships. But unfortunately that’s not immediately possible. Recall Definition ?? that $\mathcal{L}_2$ only includes functions that are square-integrable (have finite norm). However, there are some density functions for which $\int p(z) dz$ may blow up.

Example 2.1. The function $p(z) = \frac{1(0 < z < 1)}{2\sqrt{z}}$ is a valid density because it is positive and

integrates to one:

$$\frac{1}{2} \int_0^1 \frac{dz}{\sqrt{z}} = 1.$$

But it is not square-integrable: the integral of its square is

$$\frac{1}{4} \int_0^1 \frac{dz}{z} = \infty.$$

Warning

We will be drawing on properties of $\mathcal{L}_2$ substantially from here on out so this is a good time to work through Section ?? if you haven't already. At a minimum you should understand the content of Section ?? so you can understand functions as vectors.

Thankfully, this is an easy problem to fix! Instead of working with the density functions directly, we can work with the *root densities*. Since $\int p \, dz = 1$, it's clear that $\int (\sqrt{p})^2 \, dz = 1$. In other words, the square root of any density function is square-integrable and thus a function in $\mathcal{L}_2$. Moreover, all of these root densities have unit norm in $\mathcal{L}_2$! I'll use the symbol $\sqrt{\mathcal{P}} = \{\sqrt{p} : p \in \mathcal{P}\}$ to denote the collection of square roots of densities in the model. Because $\sqrt{\mathcal{P}}$ contains only positively valued functions with unit norm, we can visualize it as some subset of the unit ball in the “positive orthant” of $\mathcal{L}_2$. In describing “orthants” I am drawing on the intuition described in Section ?? where I imagine the “axes” of $\mathcal{L}_2$ corresponding to the possible arguments of the function.

2.3.1 Paths

In addition to the model, in Figure ?? you can also see what we're going to call a *differentiable path* through the model. A path (also called a one-dimensional submodel) is intuitively what it sounds like: if you were standing at a point p (or $\mathbb{P}$) in the model and you started walking in some direction, you'd be tracing out an example of a path. Formally, a path is a one-dimensional set of distributions in the model, parameterized by some $\epsilon \in \mathbb{R}$. By “parameterized” I mean that every distribution in this set can be assigned a unique number ϵ . If you consider p_0 (or $\mathbb{P}_0$) to be the “start” of the path, you can think of the parameter ϵ as encoding “how far” you are along the path (and perhaps in which direction). The density (or distribution) indexed by the particular number ϵ is often written p_ϵ (or $\mathbb{P}_\epsilon$). Paths are going to be very useful tools for us to define what we mean by distributions getting close to each other in a sense that is required for a general definition of regularity. Paths will also come up in many other places so it is worth taking some time to understand them here.

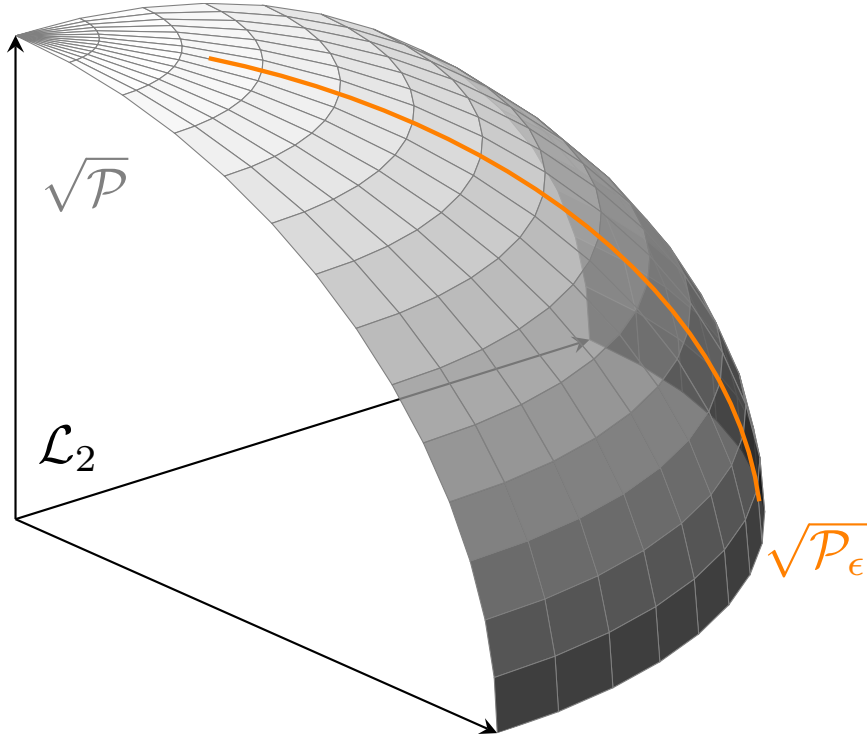


Figure 2.4: The model $\sqrt{\mathcal{P}}$ embedded in $\mathcal{L}_2$. In the drawing $\mathcal{L}_2$ is represented as 3D space but this is not meant to be literal- this is a 3D cartoon representing infinite-dimensional space. The model is represented as the visible quarter section of the unit sphere. That part of the unit sphere represents all densities. If the model were restricted in some way (e.g. all zero-mean distributions), the model could be visualized as some subset of the shown surface.

Path (1D Submodel)

Definition 2.2. A *path* is any set of distributions $\mathcal{P}_\epsilon \subseteq \mathcal{P}$ that can be put into 1-to-1 correspondence with any interval of the reals containing 0.

Example 2.2. Presume the model is the set of all continuous distributions. Pick any two points $p_0, p_1 \in \mathcal{P}$. Then the set of densities $\{(\frac{1}{2} - \epsilon)p_0 + (\frac{1}{2} + \epsilon)p_1 : \epsilon \in [0, 1]\}$. This is an example of a *convex* path. You can think of this as the straight-line path starting at p_0 and ending at p_1 (at $\epsilon = 1/2$). Some algebra shows that another way to write the elements of this path is $p_\epsilon = p_0 + \epsilon(p_1 - p_0)$.

Example 2.3. Presume the model includes all normal distributions. Then the set of distributions corresponding to $N(\epsilon, 1)$ is a path parameterized by $\epsilon \in \mathbb{R}$.

Exercise 2.4. Presume the model is the set of all mean-zero random variables. Why is $\{N(\epsilon, 1) : \epsilon \in \mathbb{R}\}$ not a path here?

Example 2.4. Presume the model is the set of all densities. For $\epsilon \leq 0$ let p_ϵ be the density of $N(\epsilon, 1)$ and for $0 < \epsilon$ let p_ϵ be the density $\text{Unif}(0, 1 + \epsilon)$.

It's important that every element of the path is an element of the model. A path, by definition, stays within the model. Later we will see that when we have a smaller (more restricted) model we can sometimes construct estimators that have smaller asymptotic variance. This should make sense to you: the more you already know and can encode as a modeling assumption, the less you have to rely on the data and the more confident you can be in your result. Mathematically, the way that we show this is by arguing that there are “fewer” paths passing through a given point when the model is more restricted. If paths were allowed to “leave” the model, this wouldn't be the case.

Our definition of path is very general. Too general, in fact, to correspond nicely to the mental model we have from Figure ???. As Example ??? shows, we can easily construct paths that don't feel “continuous”. To make a definition that reflects our picture of a continuous, smooth path through the model we need to impose more conditions.

For example, we might impose that paths be continuous in the sense that we would want the $\mathcal{L}_2$ distance between two (root) distributions $\sqrt{p_0}$ and $\sqrt{p_\epsilon}$ along the path to go to zero if $\epsilon \rightarrow \epsilon_0$. That would rule out the path in Example ???. But it turns out that what we need in order to generate a useful theory of efficiency is not just continuity, but also differentiability. That is, we are interested in continuous paths, but specifically those that do not have any “kinks”. By that we mean that we're interested in *differentiable* paths. Since we're talking about a path through infinite-dimensional space, let's spend a little time developing our intuition about derivatives before we define a differentiable path.

2.3.2 Scores

Given a function $f : \mathbb{R} \rightarrow \mathbb{R}$, recall that its derivative at the point $x = 0$ (if it exists) is defined as $\dot{f}_0 = \lim_{x \rightarrow 0} \frac{f(0) - f(x)}{x}$. Another way to write this is that there exists a number $\dot{f}_0$ such that

$$\left| \frac{f(0) - f(x)}{x} - \dot{f}_0 \right| \xrightarrow{x \rightarrow 0} 0.$$

By $x \rightarrow 0$ we mean that the statement holds when replaced by *any* sequence x_n which converges to zero.

The derivative is usually thought of the “slope” of the function f at the point in question. You can think of it also as the “direction” that f is going at that point: in the simplest interpretation, a positive derivative means that f is increasing (going up), a negative derivative means that it’s decreasing (going down). The magnitude of the derivative gives the rate of increase or decrease.

Let’s generalize this a bit. You should recognize that the value of the derivative $\dot{f}_0$ is a number specifically because the function f outputs real numbers. If f returned finite-dimensional vectors, for example, then the numerator in the “slope” fraction $\frac{f(0) - f(x)}{x}$ would also be a vector and the derivative $\dot{f}_0$ would also have to be a vector in order for the difference of the two to be defined. Now that we’re working with the difference of two vectors instead of the difference of two numbers, the appropriate measure of distance is a norm on the vector space rather than the absolute value (which is the usual norm on $\mathbb{R}$). To formalize this, let $\mathcal{V}$ be a vector space with a norm $\|\cdot\|_{\mathcal{V}}$. We’ll say that $f : \mathbb{R} \rightarrow \mathcal{V}$ is differentiable at zero if there is some element $\dot{f}_0 \in \mathcal{V}$ such that

$$\left\| \frac{f(0) - f(x)}{x} - \dot{f}_0 \right\|_{\mathcal{V}} \xrightarrow{x \rightarrow 0} 0.$$

Paths, as we have defined them (Definition ??), can be thought of as functions mapping a value $\epsilon \in \mathbb{R}$ to a (root) density $\sqrt{p_\epsilon} \in \mathcal{L}_2$. We can therefore apply the above reasoning, letting ϵ play the role of x , letting the mapping $\epsilon \mapsto \sqrt{p_\epsilon}$ play the role of f , and letting $\mathcal{L}_2$ play the role of the generic vector space $\mathcal{V}$. Since in this case f maps numbers to $\mathcal{L}_2$, the derivative of f will also be an element of $\mathcal{L}_2$, i.e. a square-integrable function.

Differentiable in Quadratic Mean (DQM), Score Function

Definition 2.3. A path $\mathcal{P}_\epsilon$ is *differentiable in quadratic mean* (or just *differentiable* for short) at $\epsilon = 0$ if there is a function $\dot{\sqrt{p_\epsilon}} \in \mathcal{L}_2$ such that

$$\left\| \frac{\sqrt{p_\epsilon} - \sqrt{p_0}}{\epsilon} - \dot{\sqrt{p_\epsilon}} \right\|_{\mathcal{L}_2} \xrightarrow{\epsilon \rightarrow 0} 0.$$

We define $s = 2 \frac{\dot{\sqrt{p_\epsilon}}}{\sqrt{p_0}}$ to be the *score function* of the path $\mathcal{P}_\epsilon$ at the point $\epsilon = 0$. Often the score can be written as the derivative of the log-likelihood evaluated at zero:

$$s(z) = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \log p_\epsilon(z).$$

The score function for a path is always in $\mathcal{L}_2(\mathbb{P}_0)$ and always mean-zero at $\mathbb{P}_0$ in the sense that $\mathbb{P}_0 s = \int s(z) p_0(z) dz = 0$.

Sometimes you may also see the DQM condition written out more explicitly and with the score function, e.g.

$$\int \left[\frac{\sqrt{p_\epsilon(z)} - \sqrt{p_0(z)}}{\epsilon} - \frac{1}{2} s(z) \sqrt{p_0(z)} \right]^2 dz \xrightarrow{\epsilon \rightarrow 0} 0.$$

This is equivalent but in my opinion makes it opaque that $\frac{1}{2} s(z) \sqrt{p_0(z)}$ just represents the $\mathbb{R} \rightarrow \mathcal{L}_2$ derivative of $\epsilon \mapsto \sqrt{p_\epsilon}$.

Proof of Claims about Scores

Proof that $s \in \mathcal{L}_2(\mathbb{P}_0)$: By definition, $\dot{\sqrt{p_\epsilon}} \in \mathcal{L}_2$ and $\dot{\sqrt{p_\epsilon}} = \frac{1}{2} s \sqrt{p_0}$. Then

$$\|s\|_{\mathcal{L}_2(\mathbb{P}_0)}^2 = \int s^2(z) p_0(z) dz = \int \left(\dot{\sqrt{p_\epsilon}}(z) \right)^2 dz = \left\| \dot{\sqrt{p_\epsilon}} \right\|_{\mathcal{L}_2}^2 < \infty.$$

Proof that $\mathbb{P}_0 s = 0$: For this proof to make sense you'll need to have worked through Section ??.

The definition of DQM is the same as saying that $\epsilon^{-1}(\sqrt{p_\epsilon} - \sqrt{p_0}) \xrightarrow{\mathcal{L}_2} \frac{1}{2} s \sqrt{p_0}$ as $\epsilon \rightarrow 0$ in the sense of Definition ??. Since ϵ^{-1} diverges, there is no way for that series to converge unless $\sqrt{p_\epsilon} \xrightarrow{\epsilon \rightarrow 0} \sqrt{p_0}$. This is a rather formal way of saying that if the path $\epsilon \mapsto \sqrt{p_\epsilon}$ is differentiable, then it certainly must be continuous since that is a weaker condition.

As a consequence of that continuity, $\sqrt{p_\epsilon} + \sqrt{p_0} \rightarrow 2\sqrt{p_0}$. Now we write

$$\mathbb{P}_0 s = \int s(z) p_0(z) dz = \int \left(\frac{1}{2} s(z) \sqrt{p_0(z)} \right) (2\sqrt{p_0(z)}) dz,$$

i.e. the $\mathcal{L}_2$ inner product between $\frac{1}{2} s \sqrt{p_0}$ and $2\sqrt{p_0}$. Notice that these two arguments are the limits of $\epsilon^{-1}(\sqrt{p_\epsilon} - \sqrt{p_0})$ and $\sqrt{p_\epsilon} + \sqrt{p_0}$ respectively because of what we showed above. Since the inner product is continuous (see Example ??), the inner product at the limits is the same as the limit of the inner products at the sequences, i.e.

$$\begin{aligned} \left\langle \frac{1}{2} s \sqrt{p_0}, 2\sqrt{p_0} \right\rangle &= \lim_{\epsilon \rightarrow 0} \epsilon^{-1} \langle \sqrt{p_\epsilon} - \sqrt{p_0}, \sqrt{p_\epsilon} + \sqrt{p_0} \rangle \\ &= \lim_{\epsilon \rightarrow 0} \epsilon^{-1} (\|\sqrt{p_\epsilon}\| - \|\sqrt{p_0}\|) \end{aligned}$$

For any density p , we already showed that $\|\sqrt{p}\| = 1$, i.e. the root-densities are on the unit ball. Thus the difference of norms on the right-hand side cancels and we get 0 as desired.

For those of you who know measure theory, you can check that this definition does not actually require a global dominating measure for the entire path which is used to define the densities. For each sequence $\epsilon_n \rightarrow 0$ needed to check the DQM criterion you may construct a bespoke dominating measure using the convex combination of the countable number of distributions $\mathbb{P}_{\epsilon_n}$.

Instead of demanding that $\epsilon \mapsto \sqrt{p_\epsilon}$ be differentiable in the mean-squared sense at $\epsilon = 0$, we might instead consider asking for the existence of the derivative $\frac{d}{d\epsilon} \sqrt{p_\epsilon(z)}$ at every single point z . When both the z -pointwise derivative $\frac{d}{d\epsilon} \sqrt{p_\epsilon(z)}$ and the mean-square derivative $\dot{\sqrt{p_\epsilon}}$ exist they certainly agree with each other, but sometimes one exists and not the other. There are paths that are not z -pointwise differentiable but are still DQM: since the square-mean derivative integrates over the squared difference of the finite- ϵ “slope” term and the limit term in z , it does not care if that slope term blows up for some points z : they simply fall out of the integral. Similarly there are z -pointwise differentiable paths that are not DQM: refer to the results and examples in Section 7.2 of Van der Vaart (2000) if you are curious to learn more. The reason we care about DQM is because it will turn out later that this is precisely the condition required to establish something called local asymptotic normality which we will use to prove a very important result.

Even though DQM is the condition we need, it’s still informative to consider the pointwise derivative because it explains why we write the derivative as $\dot{\sqrt{p_\epsilon}} = \frac{1}{2}s\sqrt{p_0}$ using the score function s . Starting with the pointwise derivative and using the chain rule,

$$\frac{d}{d\epsilon} \sqrt{p_\epsilon(z)} = \frac{1}{2}(p_\epsilon(z))^{-1/2} \frac{d}{d\epsilon} p_\epsilon(z) = \frac{1}{2} \sqrt{p_\epsilon(z)} \frac{d}{d\epsilon} \log p_\epsilon(z)$$

where in the last line we used $\frac{d}{d\epsilon} \log p_\epsilon(z) = \frac{1}{p_\epsilon(z)} \frac{d}{d\epsilon} p_\epsilon(z)$. Therefore, when the pointwise derivative exists, the score function s is equal to the rate of change of the log-likelihood at $\epsilon = 0$. For almost all practical purposes you can take this to be the definition of the score for a path at p_0 , i.e. $s(z) = \frac{d}{d\epsilon} \log p_\epsilon(z)|_{\epsilon=0}$. From a pointwise perspective, the score is telling you how much to increase or decrease the value of the (log) density at each value z in order to move from one density in the path to another. This is shown in Figure ??.

I find it most helpful to think of the score function at a point p_0 in the path as the “direction” that the path is moving when it passes through that point. You can see this is the case geometrically in Figure ??. In the figure I’ve drawn $s\sqrt{p_0}$ as the *tangent vector* to the path at the point $\sqrt{p_0}$. This is justified by the fact that $\frac{1}{2}s\sqrt{p_0}$ is the derivative of $\epsilon \mapsto \sqrt{p_\epsilon}$ and therefore encodes the “direction” that function (path) is going. Another way of seeing it is

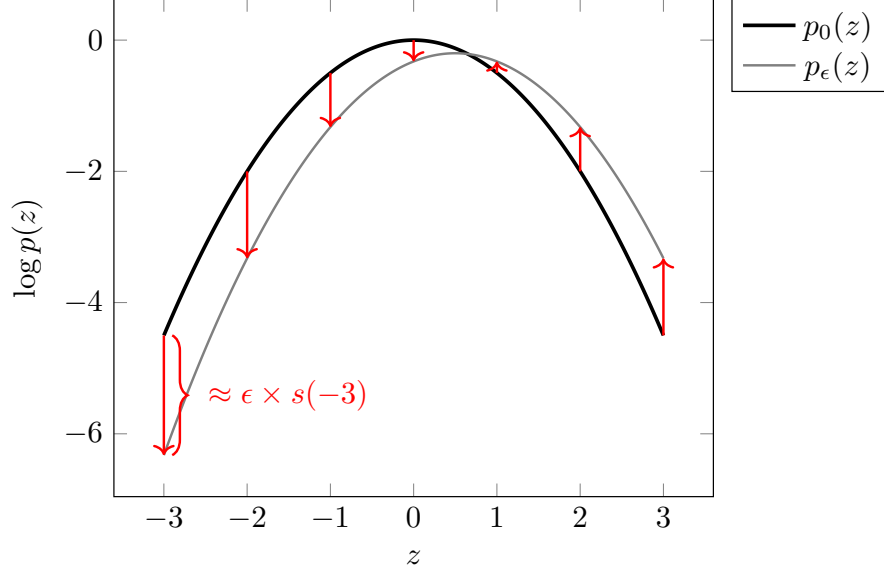


Figure 2.5: The (log-)density at ϵ for any value z can be approximated by adding the score (times ϵ) to the starting (log-)density.

that $\frac{1}{2}s\sqrt{p_0}$ is the “slope” of the best linear approximation to $\epsilon \mapsto \sqrt{p_\epsilon}$ at the point $\sqrt{p_0}$. So, roughly speaking,

$$\sqrt{p_\epsilon} \approx \sqrt{p_0} + \frac{1}{2}\epsilon(s\sqrt{p_0}).$$

This is exactly what I’ve drawn in Figure ??: the actual path of densities $\sqrt{p_\epsilon}$ (the left-hand side of the above display) and what would happen if you followed the current direction $s\sqrt{p}$ starting at $\sqrt{p}$ (the right-hand side).

To make an analogy, let’s say you were following a path from New York to San Francisco. A “path” would be equivalent to a line on the map indicating the route you take. Your “score” at some point along the route would be the direction that the route tells you to go in at that point (e.g. in New York your “score” might be “West”).

Technically it’s not the scores themselves that are tangent to paths in the model, but the scores times the current root-density that are tangent to the root-path. Nonetheless we often speak informally and talk about the scores themselves being tangent, etc, ignoring the fact that we had to take roots of the densities in our model to get everything embedded into $\mathcal{L}_2$. In line with that oversimplification, I sometimes find it helpful to think in terms of or refer to cartoons like Figure ??.

Example 2.5. Fix an initial point p_- and final point p_+ in a model and consider the convex path $\{p_0 + \epsilon(p_1 - p_0) : \epsilon \in [0, 1]\}$ discussed in Example ??. For this path, the (pointwise

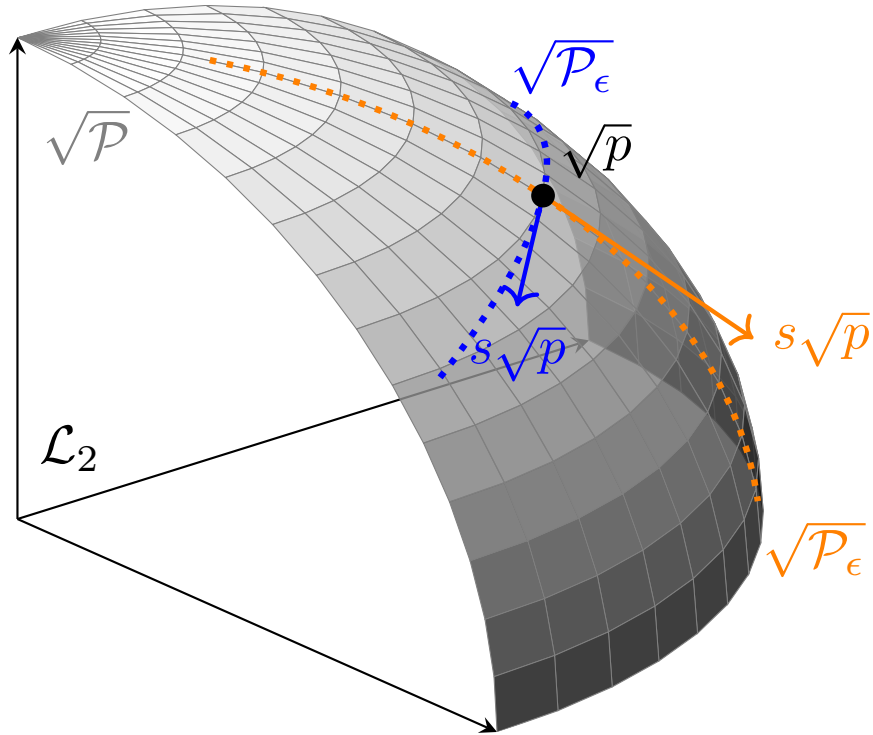


Figure 2.6: The score s (times $\sqrt{p}$) is tangent to the path $\sqrt{\mathcal{P}_\epsilon}$ at $\sqrt{p_0}$. In the figure I've shown two different paths passing through the same point $\sqrt{p}$ (in two different colors) and their respective scores.

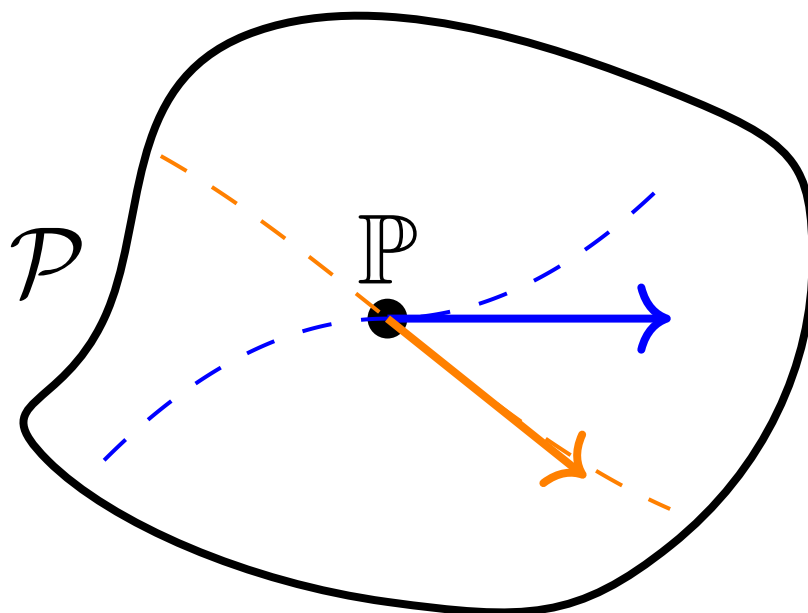


Figure 2.7: A cartoon of the model and a distribution $\mathbb{P}$ within it. Two differentiable paths are depicted with dotted lines (in different colors) and the scores for those two paths at $\mathbb{P}$ are shown as tangent vectors in the corresponding colors.

derivative version of the) score at $\epsilon = 0$ is defined and is equal to

$$\begin{aligned}
s &= \left. \frac{d}{d\epsilon} \log(p_0 + \epsilon(p_1 - p_0)) \right|_{\epsilon=0} \\
&= \left. \frac{1}{p_0 + \epsilon(p_1 - p_0)} (p_1 - p_0) \right|_{\epsilon=0} \\
&= \frac{p_1 - p_0}{p_0} \\
&= \left(\frac{p_1}{p_0} - 1 \right)
\end{aligned}$$

A little algebra shows that this path can also be written as $p_\epsilon = (1 + \epsilon s) p_0$.

Exercise 2.5. While it's most helpful to think of scores as encoding “direction” within a model, scores also encode the “speed” at which a path goes.

Presume that the score for a path $\epsilon \mapsto p_\epsilon$ is s . In terms of s , what is the score for the reparametrized path $\epsilon \mapsto p_{c\epsilon}$ for some $c \in \mathbb{R}$? To solve this problem you can use the pointwise derivative version of the score for convenience. For intuition, note that for $c > 1$, the reparametrized path is the same as the original, just traversed “faster”, whereas for $c \in (0, 1)$ it is the same path traversed more “slowly” (and for $c < 0$, in reverse).

Paths and scores are extremely useful objects in nonparametric asymptotics and we will use them many times in the rest of the book.

2.4 Regularity

Equipped with these tools we can craft a definition of regularity that is more general than Definition ???. In this section I'll return using superscripts to denote a sequence of distributions so that the subscript n can indicate the empirical distribution. For example, in the context of a path, I'll use $\mathbb{P}^{(\epsilon)}$ rather than $\mathbb{P}_\epsilon$ (and I'll use distributions $\mathbb{P}$ rather than densities p since we can now be more general).

First let's generalize the model. In our previous definition we had the normal model $\mathcal{P} = \{\mathcal{N}(\psi, 1) : \psi \in \mathbb{R}\}$. This is actually one example of a differentiable path within a larger, possibly nonparametric model (e.g. all continuous distributions). So let's replace the normal model with a generic differentiable path that includes the true distribution (at $\mathbb{P}_0$ so $\epsilon = 0$ is the true value, for convenience). Next, we can replace the condition that $\sqrt{n}\psi_n$ stays bounded with an equivalent condition on the parameter in the path. That yields our general pathwise definition of regularity:

Pathwise Regularity

Definition 2.4. An estimator $\hat{\Psi}$ is *regular* at $\mathbb{P}_0$ in the differentiable path $\mathcal{P}_\epsilon$ if there is a single, fixed distribution $\mathbb{D}$ such that

$$\sqrt{n} \left(\hat{\Psi}(\mathbb{P}_n^{(n)}) - \Psi(\mathbb{P}^{(n)}) \right) \rightsquigarrow \mathbb{D}$$

for *any* sequence of distributions $\mathbb{P}^{(n)} = \mathbb{P}^{(\epsilon_n)}$ such that $\sqrt{n}\epsilon_n$ remains bounded.

In other words, an estimator is regular in a given differentiable path if the limiting distribution doesn't depend on what distributions you draw from asymptotically as long as they move to true distribution quickly enough. Recall that $\sqrt{n}\epsilon_n$ staying bounded (not diverging) means that $\epsilon_n \rightarrow 0$ faster than the $\sqrt{n}$ factor blows up; another way to write this condition is $\epsilon_n = O(n^{-1/2})$. The “uniform” behavior of regular estimators across “local” distributions in the path is what prevents the bad behavior of Hodges' estimator. The defects of irregularity escape detection in pointwise asymptotic arguments where the distribution is fixed.

To generalize our definition to a larger, possibly nonparametric model, we simply require that the limiting distribution $\mathbb{D}$ is also the same no matter which differentiable path through $\mathbb{P}$ you choose. In other words, we require the “uniformity” of convergence in distribution across all sequences of distributions converging quickly to the truth along differentiable paths.

Regularity

Definition 2.5. An estimator $\hat{\Psi}$ is *regular* at a point $\mathbb{P}$ in the model $\mathcal{P}$ if it is regular with the same limiting distribution $\mathbb{D}$ along all paths $\mathcal{P}_\epsilon \subseteq \mathcal{P}$ that are differentiable at $\mathbb{P}_0 = \mathbb{P}$. An estimator is regular in a model $\mathbb{P}$ if it is regular at every point $\mathbb{P} \in \mathcal{P}$.

We do not care about the behavior of the estimator along non-differentiable paths because these paths don't go smoothly towards the true distribution. It would not be sensible to assume that any estimator should behave in a predictable or consistent way if we moved along a non-differentiable path. Moreover we do not care about sequences of distributions along paths that do not converge quickly enough (at a $n^{-1/2}$ rate) to zero: we have no need to impose uniformity outside of “neighborhoods” of distributions that are roughly of “radius” $n^{-1/2}$ in order to get a nice, general theory of optimality.

Proving that an estimator is not regular is only as difficult as finding a single differentiable path for which it is not regular. We already did that for Hodges' estimator in Exercise ???. To directly prove an estimator *is* regular you have to prove regularity for all differentiable paths. There are ways to do this but more often it is easier to prove regularity *indirectly*. In the next chapter we will prove a powerful result that lets us characterize the set of all regular, asymptotically linear estimators. We can then show an estimator is regular by proving it is asymptotically linear and also meets the conditions required for membership in that set. More

importantly, the characterization we develop will let us precisely identify the *most efficient* regular and asymptotically linear estimator.

2.5 Is Regularity Too Strict?

By restricting our attention to *regular* asymptotically linear estimators we are trying rule out the kind of bad performance exhibited by Hodges’ estimator. But Hodges’ estimator is just one example. It’s rational to ask whether there are any *good* estimators that regularity rules out. In practice, the short answer is no, but in theory the long answer is that it depends on your perspective. I won’t give a detailed account of all the arguments but I will give a brief summary that should at a minimum hold you over until you can dig deeper. Nonetheless, if you are interested in these questions I recommend revisiting them *after* you have digested the practical implications of efficiency theory that we’ll discuss in the next chapters.

The first argument that usually gets rolled out in favor of enforcing regularity is that irregular estimators can only perform asymptotically better than regular estimators on an “infinitesimally small” fraction of all possible distributions in any parametric model. We saw this for Hodges’ estimator: it has the same asymptotic performance as the sample mean in the normal model everywhere except for the *single distribution* where $\psi = 0$. That’s the only point at which Hodges’ estimator can beat the sample mean in large-enough samples. The theoretical justification for this claim is something called the almost-everywhere (or Hajek-Le Cam) convolution theorem. You can see section 8.6 of Van der Vaart (2000) for further details. The idea is that if irregular estimators can only beat regular estimators in very rare cases, it’s no big deal to exclude them. As a bonus, the convolution theorem also shows that the most efficient regular estimator is asymptotically linear. This justifies restricting our search to asymptotically linear estimators once we’re already comfortable with regularity since we lose nothing (asymptotically) by the further restriction.

A second argument for regularity follows from the local asymptotic minimax theorem (Theorem 8.11 and Lemma 8.13 in Van der Vaart (2000)). Given an estimate $\hat{\psi}$, we can imagine calculating a scaled error $E_{n,\mathbb{P},\hat{\Psi}} = \sqrt{n}(\hat{\psi} - \psi)$. This error is indexed by the sample size n , the distribution $\mathbb{P}$ we’re drawing data from and calculating the estimand ψ from, and of course the estimator $\hat{\Psi}$. Since $\hat{\psi}$ is random, $E_{n,\mathbb{P},\hat{\Psi}}$ is also a random variable: we get a different error with a different sampled dataset. We can also define a loss function $l : \mathbb{R} \rightarrow \mathbb{R}$ which takes the error and returns its “cost”. We’ll require that the loss be “bowl-shaped”: symmetric around 0 and either increasing or flat moving away from zero. For example, we might use the squared loss function $l(e) = e^2$ if we want to strongly penalize large errors but be relatively more lenient towards small errors, or we could use $l(e) = 1(|e| > c)$ for a constant penalty outside a region and no penalty within it.

The local asymptotic minimax theorem is concerned with the expected loss $\mathbb{E}_{\mathbb{P}}[l(E_{n,\mathbb{P},\hat{\Psi}})]$. Clearly we would like to find an estimator that makes the expected loss as small as possible. But the theorem tells us that there is a limit to how low the expected loss can get if

we want the loss to be *uniformly* low in some sense around a distribution $\mathbb{P}$. Let $\mathbb{P}^{(n)}$ denote any sequence $\mathbb{P}^{(\epsilon_n)}$ in a differentiable path with $\mathbb{P}_0 = \mathbb{P}$ such that $\sqrt{n}\epsilon \rightarrow 0$. Informally, the theorem establishes that in most models, if we consider a large enough set of possible differentiable paths and sequences and take n to be large enough, then there is *no* estimator for which the worst-case value (over paths) of $\mathbb{E}_{\mathbb{P}^{(n)}}[l(E_{n,\mathbb{P}^{(n)},\hat{\Psi}})]$ can be meaningfully better than $\mathbb{E}_{Z \sim \mathcal{N}(0, \sigma_{\mathbb{P}}^2)}[l(Z)]$. The quantity $\mathbb{E}_{Z \sim \mathcal{N}(0, \sigma_{\mathbb{P}}^2)}[l(Z)]$ is a sort of universal “speed limit” for good locally-uniform estimation of the estimand at $\mathbb{P}$ as measured by the loss l . In the next few chapters we’ll say a lot more about the mysterious quantity $\sigma_{\mathbb{P}}^2$ and how to derive it.

It’s natural to ask whether there are estimators that can meet this bound, i.e. whether there are estimators that are *locally asymptotically minimax* for a loss l . If such an estimator were available, there would be very good reason to prefer it over alternatives. The second part of the local asymptotic minimax theorem establishes that, roughly speaking, only *regular* estimators can be locally asymptotically minimax for scalar estimands. One implication of this theorem is that the undesirable behavior exhibited by Hodges’ estimator is characteristic of *all* superefficient estimators. If there is no *irregular* locally asymptotically minimax estimator, that means that irregular estimators *all* have regions of bad performance with in the model that shift towards the point of superefficiency as n increases.

Together, the convolution and local asymptotic minimax theorem provide strong reasons to prefer regular estimators, but ultimately both are asymptotic statements and say nothing about how irregular estimators might perform in finite samples. Moreover, parts of the local asymptotic minimax theorem don’t hold if the estimand is a vector rather than a scalar. For example, the James-Stein estimator is an *irregular* estimator of a vector-valued mean that *also* locally asymptotically minimax (in a sense defined for vector estimands) for the squared error loss (see Section 8.8 of Van der Vaart (2000) or Chapter 2 of Ibragimov and Has’ Minskii (2013)). But even this is a qualified improvement: while James-Stein improves on the sample mean for squared error loss, there are other losses for which it can be much worse.

Ultimately, the strongest argument in favor of assuming regularity is a practical one: if you assume regularity, you get a very clear theory of optimality that works in immense generality for many statistical models and estimands. It’s possible to cook up small-sample scenarios where irregular estimators can win, but given a general setup and without very specific assumptions it’s not usually clear how to *generate* a good irregular estimator from scratch. Moreover it can be difficult to characterize the sampling distribution of irregular estimators, making it hard to construct confidence intervals and p-values. In contrast, our efficiency theory for regular estimators will allow us to build great estimators with easy ways to do inference for essentially any situation.

3 Efficiency for RAL Estimators

In Chapter ?? we defined *asymptotic linearity* (AL) and discussed why AL estimators are nice: they are asymptotically unbiased, are easy to do inference for, and can be (asymptotically) compared by looking at their influence functions. In Chapter ?? we argued that we should further restrict the class of desirable estimators to those that are *regular* so that bad finite-sample behavior can't be hidden in the asymptotic comparisons. In this chapter we will establish the central result of efficiency theory: in any model where the estimand varies smoothly enough over the different distributions, there is a best-possible regular asymptotically linear (RAL) estimator.

Really what we'll prove is that there is a best-possible *influence function*: multiple estimators may have that influence function but by virtue of that fact they will behave similarly in large-enough samples. We call the best-possible influence function the *efficient* influence function (EIF). The EIF sets a limit on how good any RAL estimator can be at estimating some estimand in a some model. If you have a RAL estimator with that influence function, you can rest assured that it is the best you can do (asymptotically).

In this chapter we will prove our central result and discuss some important implications. In the next chapter we'll show how to derive the EIF in the most common scenarios. After that, in the last part of the book, we will demonstrate how to use knowledge of the EIF to *construct* efficient estimators.

3.1 Characterizing Influence Functions

Not every possible function corresponds to a valid RAL estimator of an estimand in a particular statistical model. Clearly it would be wonderful if we could find a RAL estimator with $\phi(z) = 0$ because this estimator would have no variance at all in large samples. But if that were possible, statistics would be easy. So $\phi(z) = 0$ must not be an influence function for any sensible estimand in any sensible model. So, for a given estimand and statistical model, what functions are influence functions of RAL estimators, and what functions aren't?

We already know that any regular estimator by definition behaves in a particular way asymptotically along a quickly-converging sequence of distributions. Remarkably, it turns out that asymptotic linearity also implies specific asymptotic behavior along a subset of these sequences.

Asymptotic linearity along differentiable paths

Theorem 3.1. *Let $\mathcal{P}_\epsilon \subseteq \mathcal{P}$ be any differentiable path with score function s and let $\mathbb{P}^{(\epsilon_n)}$ be a sequence of distributions along this path with $\epsilon_n = c_n n^{-1/2}$ for any converging sequence of real numbers $c_n \rightarrow c$. Let $\mathbb{P} = \mathbb{P}^{(0)}$ denote the true data-generating distribution. If an estimator $\hat{\Psi}$ is asymptotically linear with influence function ϕ , then*

$$\sqrt{n} \left(\hat{\Psi}(\mathbb{P}_n^{(\epsilon_n)}) - \Psi(\mathbb{P}) \right) \rightsquigarrow \mathbf{N}(\mathbb{P}[s\phi], \mathbb{P}\phi^2).$$

Proof Outline

A thorough proof of this result is too technical for this book so I will only give an outline here. While I do encourage you to dig into the theory if you are curious (since it has other applications as well), I think this is best done *after* understanding the implications of the result in our present context.

Let p and p_{ϵ_n} denote the densities of $\mathbb{P} = \mathbb{P}^{(0)}$ and $\mathbb{P}^{(\epsilon_n)}$, respectively (you can either assume we're working in a model with densities or recall that we can always construct densities along sequences of a differentiable path).

The first step is to show that any differentiable path is in fact a *locally asymptotically normal* (LAN) model. This is proved as Theorem 7.2 in Van der Vaart (2000) and while the arguments are accessible with basic asymptotic statistics, I don't think they add too much insight. In LAN models we only consider sequences of the type $\epsilon_n = c_n n^{-1/2}$ where $c_n \rightarrow 1$. This kind of sequence is a special case of the sequences where $\sqrt{n}\epsilon_n$ stays bounded that we considered in the definition of regularity. The reason we need to consider this special case is that these are the sequences for which $\sqrt{n}(\sqrt{p_{\epsilon_n}} - \sqrt{p})$ converges in $\mathcal{L}_2$ to a fixed limit $\frac{1}{2}s\sqrt{p}$ (by Definition ??). The $\sqrt{n}$ convergence of the root densities is required to show that

$$\sum_i^n \log \frac{p_{\epsilon_n}(Z_i)}{p(Z_i)} = \frac{1}{\sqrt{n}} \sum_i^n s(Z_i) - \frac{1}{2} \mathbb{E}[s(Z)^2] + o_{\mathbb{P}}(1)$$

along such sequences. When the above holds for a model along such sequences, we say the model is LAN.

The LAN property has nothing to do with any estimator, it is purely a statement about what the densities of nearby distributions p_{ϵ_n} look like relative to p as we sample more data from them and move closer to p . The log density ratio shown above is actually exactly of the form you get if you instead look at the log density ratio of two normal distributions with the same variance but different means. That's why the property is called local asymptotic normality. So the result says that as you “zoom in” to a point on a differentiable path while sampling more data, the model looks more and more like a set of normal distributions with different means. This happens for any path where the densities vary smoothly (differentiably).

Incredibly, local asymptotic normality tells us enough about what the densities of nearby data-generating distributions look like to infer how any asymptotically linear estimator behaves along these sequences of distributions. The implied pathwise behavior is

$$\sqrt{n} \left(\hat{\Psi}(\mathbb{P}_n^{(\epsilon_n)}) - \Psi(\mathbb{P}) \right) \rightsquigarrow \mathbf{N}(\mathbb{P}[s\phi], \mathbb{P}\phi^2)$$

as desired. The technical mechanism for inferring this is called Le Cam’s 3rd Lemma (see Example 6.7 in Van der Vaart (2000)). To prove the result in generality requires contiguity arguments (Chapter 6 of Van der Vaart (2000)), but if you assume that every distribution in the model has a density and that all the densities have the same support (set of values z for which the density is nonzero), then you can prove an equivalent result using only “basic” arguments (continuous mapping and a characteristic function argument). Either way, I don’t think the proof gives much intuition that’s relevant to our purposes so I’ve omitted it.

This is remarkable because asymptotic linearity is a pointwise property: it describes only what happens if the estimator is evaluated with more and more data sampled from the *same* distribution. By itself it says nothing about what would happen if we also *changed* that distribution while letting n grow. But if we take into account differentiability along the path suddenly we can understand that changing pathwise behavior. Moreover the pathwise behavior depends crucially on what the score s is for the path in question: notice that the mean of the limiting distribution is $\mathbb{P}s\phi$.

The benefit of knowing this is that now we can compare to the definition of regularity (Definition ??), which *also* tells us how the estimator behaves along sequences including the kind we considered in Theorem ??, but this time in a way that does *not* depend on what the score is for the path:

$$\sqrt{n} \left(\hat{\Psi}(\mathbb{P}_n^{(\epsilon_n)}) - \Psi(\mathbb{P}^{(\epsilon_n)}) \right) \rightsquigarrow \mathbb{D}.$$

Recall that $\mathbb{D}$ is some fixed distribution that by definition is not allowed to depend on the sequence or path. If we let $\epsilon_n = 0$ (a valid choice), the resulting sequence is $\sqrt{n} \left(\hat{\Psi}(\mathbb{P}_n) - \Psi(\mathbb{P}) \right)$ which converges to $\mathbf{N}(0, \mathbb{P}\phi^2)$ when the estimator is asymptotically normal (Definition ??). Since the limiting distribution $\mathbb{D}$ must be the same across all paths for the estimator to be regular at $\mathbb{P}$, it must be that $\mathbb{D} = \mathbf{N}(0, \mathbb{P}\phi^2)$.

Therefore, combining this and Theorem ??, for any estimator that is both regular *and* asymptotically linear,

$$\begin{aligned} \sqrt{n} \left(\hat{\Psi}(\mathbb{P}_n^{(\epsilon_n)}) - \Psi(\mathbb{P}) \right) &\rightsquigarrow \mathbf{N}(\mathbb{P}[s\phi], \mathbb{P}\phi^2) \\ \sqrt{n} \left(\hat{\Psi}(\mathbb{P}_n^{(\epsilon_n)}) - \Psi(\mathbb{P}^{(\epsilon_n)}) \right) &\rightsquigarrow \mathbf{N}(0, \mathbb{P}\phi^2) \end{aligned}$$

for any sequences $\mathbb{P}^{(\epsilon_n)}$ along a differentiable path with $\epsilon_n = c_n n^{-1/2}$ and $c_n \rightarrow 1$. Comparing the left- and right-hand sides above, the only way that both asymptotic statements can be true simultaneously is if

$$\sqrt{n}(\Psi(\mathbb{P}^{(\epsilon_n)}) - \Psi(\mathbb{P})) \rightarrow \mathbb{P}[s\phi].$$

Because $\epsilon_n \sim 1/\sqrt{n}$ and the choice of differentiable path was arbitrary, this implies the following result.

Influence Functions of RAL Estimators

Theorem 3.2. *If ϕ is an influence function of an estimator that is RAL at $\mathbb{P} \in \mathcal{P}$, then*

$$\lim_{\epsilon \rightarrow 0} \frac{\Psi(\mathbb{P}^{(\epsilon)}) - \Psi(\mathbb{P}^{(0)})}{\epsilon} = \mathbb{P}s\phi$$

for any differentiable path $\mathcal{P}_\epsilon = \{\mathbb{P}^{(\epsilon)} : \epsilon \in \mathbb{R}\}$ with score function s .

We often abbreviate the left-hand side of the above with the notation $\dot{\Psi}(s)$ to indicate that it is a derivative of Ψ at $\mathbb{P}$ in the “direction” s in which case the identity is written cleanly as

$$\dot{\Psi}_{\mathbb{P}}(s) = \mathbb{P}s\phi.$$

Any asymptotically linear estimator with influence function ϕ that satisfies this identity for all differentiable paths is regular at $\mathbb{P} \in \mathcal{P}$.

Proof

At first it may seem like there is nothing to prove since substituting $\sqrt{n} \approx \epsilon_n^{-1}$ appears to do the trick, but just because

$$\lim_{n \rightarrow \infty} \frac{f(\epsilon_n) - f(0)}{\epsilon_n} = \dot{f}$$

for all sequences $\epsilon_n = c_n n^{-1/2}$ with $c_n \rightarrow 1$ does not obviously imply that the same is true for *any* sequence $\epsilon_n \rightarrow 0$ (here I am abbreviating $f(\epsilon) = \Psi(\mathbb{P}^{(\epsilon)})$ and $\dot{f} = \mathbb{P}s\phi$). The latter convergence along any sequence $\epsilon_n \rightarrow 0$ is what we need for the left-hand side above to define a derivative.

To do this rigorously requires a little more effort. The rigorous proof is a bit tedious and not at all important to understand if you are happy to believe the result. I include it here only because it relies only on standard tools from analysis and I have not found a thorough version elsewhere in the literature.

In terms of the abbreviations with f , the display above Theorem ?? says that

$$\sqrt{n}[f(\epsilon_n) - f(0)] \rightarrow \dot{f}.$$

Since $\epsilon_n = c_n n^{-1/2}$, the above is equivalent to

$$c_n \epsilon_n^{-1} [f(\epsilon_n) - f(0)] \rightarrow \dot{f}.$$

Since $c_n \rightarrow 1$ this happens only if $\epsilon_n^{-1} [f(\epsilon_n) - f(0)] \rightarrow \dot{f}$. Therefore $\epsilon_n^{-1} [f(\epsilon_n) - f(0)] \rightarrow \dot{f}$ for all $\epsilon_n = c_n n^{-1/2}$ with $c_n \rightarrow 1$. However to prove the theorem we need this to hold for all $\epsilon_n \rightarrow 0$. To show this, we start by picking any sequence $\eta_m \rightarrow 0$. Our goal is to show that this can be written in a way similar to $c_n n^{-1/2}$ with $c_n \rightarrow 1$ so that the difference quotient $\eta_m^{-1} [f(\eta_m) - f(0)]$ converges for this sequence by what we've already shown.

Let $n_m = \lfloor 1/\eta_m^2 \rfloor$ be the value of $1/\eta_m^2 \geq 0$ rounded down to the nearest integer for each m . Since $\eta_m \rightarrow 0$ by assumption, n_m is a sequence of positive integers that is eventually diverging. If we define $c_m = \eta_m n_m = \eta_m \sqrt{\lfloor 1/\eta_m^2 \rfloor}$ then $|c_m| \rightarrow 1$ because the function $\sqrt{\lfloor 1/\eta^2 \rfloor}$ gets closer and closer to $\sqrt{1/\eta^2}$ as η gets smaller.

If you care to see that rigorously, let $\zeta = 1/\eta^2$, which must diverge. The desired convergence

$$\left| \sqrt{1/\eta^2} - \sqrt{\lfloor 1/\eta^2 \rfloor} \right| \xrightarrow{\eta \rightarrow 0} 0$$

is true if

$$\left| \sqrt{\zeta} - \sqrt{\lfloor \zeta \rfloor} \right| \xrightarrow{\zeta \rightarrow \infty} 0.$$

but

$$\left| \sqrt{\zeta} - \sqrt{\lfloor \zeta \rfloor} \right| = \sqrt{\zeta} - \sqrt{\lfloor \zeta \rfloor} \leq \sqrt{\lfloor \zeta \rfloor + 1} - \sqrt{\lfloor \zeta \rfloor} = \frac{1}{\sqrt{\lfloor \zeta \rfloor + 1} + \sqrt{\lfloor \zeta \rfloor}}$$

The final expression on the right converges to zero for any $\zeta \rightarrow \infty$, which establishes $|c_m| \rightarrow 1$.

We have therefore established that any sequence $\eta_m \rightarrow 0$ can be represented as a sequence $c_m/\sqrt{n_m}$ where $|c_m| \rightarrow 1$ and n_m are integers diverging to infinity. This is almost of the form we need except for the fact that 1) n_m is an arbitrary diverging sequence of integers rather than the counting sequence $\{1, 2, 3, \dots\}$ and 2) we have $|c_m| \rightarrow 1$ instead of $c_m \rightarrow 1$. To resolve this we will first assume $c_m \rightarrow 1$ and show that the convergence of the difference quotient along sequences $c_n/\sqrt{n}$ actually implies convergence along sequences $c_m/\sqrt{n_m}$ first with n_m strictly increasing and then with n_m an arbitrary diverging sequence of integers. After that is done we will address the issue of the absolute value on c_m .

For any increasing sequence n_m and sequence $c_m \rightarrow 1$, build the "supersequence" $\tilde{c}_n = c_m$ for all $n_m \leq n < n_{m+1}$ so that $c_m = \tilde{c}_{n_m}$ by construction and $\tilde{c}_n \rightarrow 1$ since $c_m \rightarrow 1$. Again by construction, $\tilde{\eta}_m = \tilde{c}_{n_m}/\sqrt{n_m}$ is a subsequence of $\epsilon_n = c_n/\sqrt{n}$ which is of the desired form. Because of that, $\tilde{\eta}_m^{-1} [f(\tilde{\eta}_m) - f(0)]$ is a subsequence of $\epsilon_n^{-1} [f(\epsilon_n) - f(0)]$ and thus must converge to the same limit $\dot{f}$.

Now let n_m be an arbitrary, diverging, positive sequence of integers and $c_m \rightarrow 1$. Since n_m is diverging, any subsequence n_{m_k} is also diverging. From any diverging subsequence we can take a strictly increasing subsequence $n_{m_{k_j}} < n_{m_{k_{j+1}}}$. Along this sub-subsequence, $c_{m_{k_j}} \rightarrow 1$. Therefore, letting $\bar{\eta}_j = c_{m_{k_j}} / \sqrt{n_{m_{k_j}}}$, we know that $\bar{\eta}_j^{-1}[f(\bar{\eta}_j) - f(0)] \rightarrow \dot{f}$ because $\bar{\eta}_j$ satisfies the conditions required from the paragraph above. Since every subsequence $\left(\frac{c_{m_k}}{\sqrt{n_{m_k}}}\right)^{-1} \left[f\left(\frac{c_{m_k}}{\sqrt{n_{m_k}}}\right) - f(0)\right]$ has a further subsequence that converges to a common limit $\dot{f}$, that means the original sequence $\left(\frac{c_m}{\sqrt{n_m}}\right)^{-1} \left[f\left(\frac{c_m}{\sqrt{n_m}}\right) - f(0)\right]$ also converges to that common limit. This is a consequence of the fact that a sequence converges if and only if all of its subsequences converge to the same limit.

Now we deal with the issue of the absolute value on the sequence c_m . To do this we consider a sequence along a path $\mathcal{P}_\epsilon$ indexed by $\epsilon_n = c_n n^{-1/2}$ with $c_n \rightarrow -1$ instead of $c_n \rightarrow 1$. This is exactly the same as a sequence with $c_n \rightarrow 1$ but along the “reversed” path $\mathcal{P}_{-\epsilon}$ which has score $-s$ (see Exercise ??). Therefore along such sequences $c_n \epsilon_n^{-1}[f(\epsilon_n) - f(0)] \rightarrow -\dot{f}$ by our starting proposition. Multiplying both sides by -1 we obtain that $(-c_n) \epsilon_n^{-1}[f(\epsilon_n) - f(0)] \rightarrow \dot{f}$ for any sequence $c_n \rightarrow -1$ and $\epsilon_n = c_n n^{-1/2}$. Since $-c_n \rightarrow 1$, by the argument at the beginning of this proof we know that $\epsilon_n^{-1}[f(\epsilon_n) - f(0)] \rightarrow \dot{f}$ for any sequence $\epsilon_n = c_n n^{-1/2}$ now for either $c_n \rightarrow 1$ or $c_n \rightarrow -1$.

We may now return to our original sequence $c_m = \eta_m n_m$ whose absolute value converges to 1. In a mild abuse of notation let c_m^+ be the subsequence of the non-negative entries of c_m and let c_m^- be the negative entries (with corresponding subsequences n_m^+ and n_m^- which are both diverging sequences of integers). If c_m^+ has a finite number of elements then $c_m \rightarrow -1$ and otherwise if c_m^- has a finite number of elements then $c_m \rightarrow 1$. Either way $\eta_m^{-1}[f(\eta_m) - f(0)] \rightarrow \dot{f}$ by the argument of the paragraph above. In the case that both c_m^+ and c_m^- are infinite sequences, $|c_m| \rightarrow 1$ implies that $c_m^+ \rightarrow 1$ and $c_m^- \rightarrow -1$. Therefore we may divide η_m into the two corresponding sequences η_m^+ and η_m^- and since $(\eta_m^\pm)^{-1}[f(\eta_m^\pm) - f(0)] \rightarrow \dot{f}$ for either part of the sequence it must be that $\eta_m^{-1}[f(\eta_m) - f(0)] \rightarrow \dot{f}$ for the entire sequence.

Lastly, the claim that any asymptotically linear estimator with influence function ϕ that satisfies $\dot{\Psi}_{\mathbb{P}}(s) = \mathbb{P}s\phi$ for all differentiable paths is automatically regular at $\mathbb{P} \in \mathcal{P}$ comes from running the argument above in reverse: if the conclusion holds and asymptotic linearity holds, then regularity must hold or else a contradiction arises (see also Proposition 1 in Chapter 3 or Bickel et al. (1993)).

At a glance it seems like the limit $\lim_{\epsilon \rightarrow 0} \frac{\Psi(\mathbb{P}(\epsilon)) - \Psi(\mathbb{P}(0))}{\epsilon}$ should depend on the particular path $\mathcal{P}_\epsilon$. If this were the case it would be more appropriate to abbreviate it with notation like $\dot{\Psi}_{\mathbb{P}}(\mathcal{P}_\epsilon)$. However, the notation $\dot{\Psi}_{\mathbb{P}}(s)$ is justified by the fact that the limit must be the same value $\mathbb{P}s\phi$ for any two paths $\mathcal{P}_\epsilon$ and $\mathcal{P}'_\epsilon$ that have the same score s . That is, the limit is not really a function of the particular path $\mathcal{P}_\epsilon$, but a function of the score s .

Theorem ?? is an astounding result because it connects two seemingly unrelated things: the

left-hand side is the rate of change in the estimand as we move the underlying distribution along a path. That has nothing to do with any estimator. The right-hand side is some measure of the “angle” between the influence function of an estimator and the direction of the path. By assuming regularity and asymptotic normality we have managed to connect these two seemingly unrelated quantities. What this means about the set of possible influence functions may still be unclear, but for the moment I hope you can appreciate the result as a very interesting connection nonetheless!

3.2 Tangent Sets

Theorem ?? holds simultaneously for *all* differentiable paths in the model through $\mathbb{P}$ (and all their induced scores). To unlock the important insights from the theorem we therefore need to be able to refer to that set of scores.

Tangent Set

Definition 3.1. The *tangent set* $\dot{\mathcal{P}}_{\mathbb{P}}$ at a point $\mathbb{P} \in \mathcal{P}$ is the set of all scores of all differentiable paths through that point. When the point of interest $\mathbb{P}$ is clear from context we leave the subscript off and just write $\dot{\mathcal{P}}$ for the tangent set.
 $\dot{\mathcal{P}}_{\mathbb{P}} \subseteq \{f \in \mathcal{L}_2(\mathbb{P}) : \mathbb{P}f = 0\} \equiv \mathcal{L}_2^0(\mathbb{P})$ since any score s of a differentiable path through $\mathbb{P}$ is in $\mathcal{L}_2(\mathbb{P})$ and has mean zero by Definition ??.

Very often the tangent set is a closed vector space (i.e. it is closed under sums, scaling, and limits; see Definition ??). When this is the case, some authors refer to the tangent set as the tangent *space*. If the tangent set is not already a closed vector space, the term “tangent space” usually refers to the linear closure of the tangent set (all linear combinations of scores and their $\mathcal{L}_2$ -limits).

Looking at Figure ??, it’s clear why we call this the “tangent” set: the plane $\dot{\mathcal{P}}_{\mathbb{P}}\sqrt{p}$ is tangent to the surface of the root-model $\sqrt{\mathcal{P}}$ at the density $\sqrt{p}$. That is also why we use the evocative dot notation for the tangent set: $\dot{\mathcal{P}}_{\mathbb{P}}\sqrt{p}$ contains all of the derivatives $\dot{\sqrt{p}}_{\epsilon}$. Although the picture in fact shows the root-model and tangent set times the root density, we often say informally that the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$ itself is “tangent” to the model $\mathcal{P}$ at the distribution $\mathbb{P}$.

Since it’s useful to think of scores as directions we can move away from $\mathbb{P}$ in the model, the tangent set is simply the set of all such directions. If a model is “large” in a neighborhood of $\mathbb{P}$ in the sense that there are many directions you can go in and stay in the model, then the tangent set will be large. If a model is “small” around $\mathbb{P}$ then the tangent set will be small.

Example 3.1. Consider the model $\mathcal{P} = \{\mathbf{N}(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}$ with densities $p_{\mu, \sigma^2}(z)$ and the point $\mathbb{P} = \mathbf{N}(0, 1)$ with density $p(z) = (2\pi)^{-1/2}e^{-z^2/2}$ of the standard normal. Since this is a two-dimensional parametric model, the tangent set at p is all linear combinations of

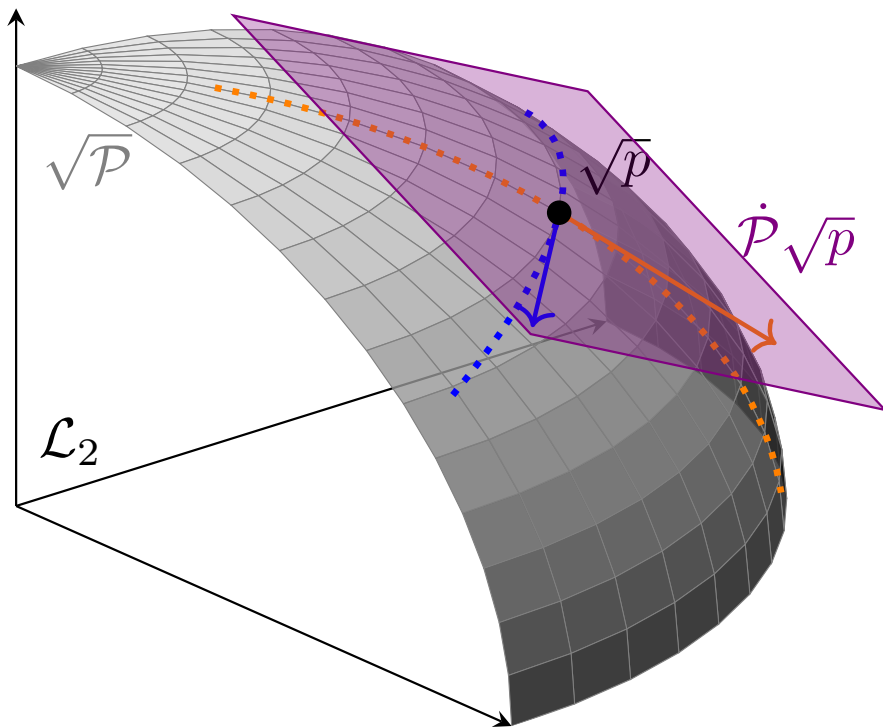


Figure 3.1: A visual representation of the tangent set $\dot{\mathcal{P}}$ along with the model $\sqrt{\mathcal{P}}$, two differentiable paths, and two scores (times $\sqrt{p}$). The set $\dot{\mathcal{P}}\sqrt{p}$ denotes the set of all scores times the root density: $\{s\sqrt{p} : s \in \dot{\mathcal{P}}\}$.

$$s_\mu(z) = \left. \frac{\partial p_{\mu, \sigma^2}(z)}{\partial \mu} \right|_{\mu=0, \sigma^2=1} = z$$

$$s_{\sigma^2}(z) = \left. \frac{\partial p_{\mu, \sigma^2}(z)}{\partial \sigma^2} \right|_{\mu=0, \sigma^2=1} = \frac{z^2 - 1}{2}$$

Thus in this case $\dot{\mathcal{P}}$ is a two-dimensional set of functions.

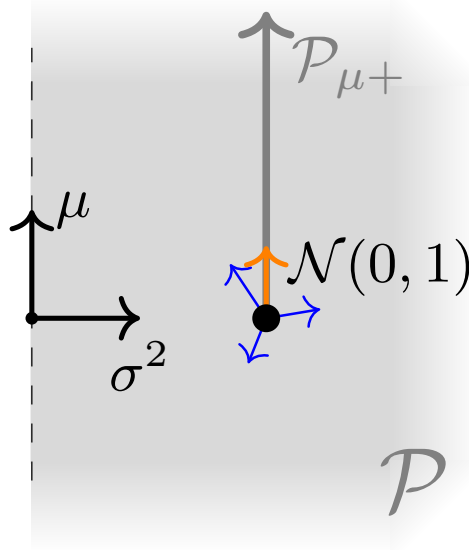


Figure 3.2: A cartoon representation of the models $\mathcal{P}$ and $\mathcal{P}_{\mu+}$ discussed in this example. The arrows emanating from $\mathbf{N}(0, 1)$ indicate scores of paths in $\mathcal{P}$, but only the orange arrow is a score of a path in $\mathcal{P}_{\mu+}$.

If we instead restricted the model to be $\mathcal{P}_\mu = \{\mathbf{N}(\mu, 1) : \mu \in \mathbb{R}\}$ by fixing $\sigma^2 = 1$ (presume we knew this a-priori), then the tangent set would be all linear combinations of s_μ alone, i.e., $\{cs_\mu : c \in \mathbb{R}\}$, a one-dimensional subset of the tangent set for the larger model.

Lastly, if we considered the even smaller model $\mathcal{P}_{\mu+} = \{\mathbf{N}(\mu, 1) : \mu \geq 0\}$, then the point $\mathbb{P} = \mathbf{N}(0, 1)$ would be on the “edge” of this model. Consequently there is no differentiable path that goes in the “negative μ ” direction from $\mathbb{P}$ and therefore the tangent set would be restricted to $\{cs_\mu : c \geq 0\}$, again a subset of the tangent space in the larger parent model. On the other hand, if we considered the point $\mathbb{P} = \mathbf{N}(1, 1)$ in this same model, the tangent space set would be $\{cs_\mu : c \in \mathbb{R}\}$, the same as in the larger model where μ is allowed to be negative. This is because $\mathbf{N}(1, 1)$ is in the “interior” of the restricted model, so the mean can be decreased without any problem.

This example demonstrates why the tangent set is specific to a particular point in a particular model. The notation $\dot{\mathcal{P}}_\mathbb{P}$ makes the model dependence very explicit and that is why I prefer

it to something more cumbersome like $\mathcal{S}_{\mathcal{P}, \mathbb{P}}$. The tangent set depends crucially on where the point is within the model because that affects what directions it is legal to move in while remaining in the model (recall that paths are subsets of the model). The shape and size of the tangent set is determined entirely by the *local* character of the model at $\mathbb{P}$. We saw in the example above that the tangent set at $N(0, 1)$ became smaller if we restricted the mean in the normal model to be non-negative, but at $N(1, 1)$ the tangent set was unaffected by that restriction. There are many more examples of specific tangent sets in Chapters 3 and 4 of Bickel et al. (1993) but for now we can proceed with the intuition we have without getting bogged down in details.

In Definition ?? we noted that $\dot{\mathcal{P}}_{\mathbb{P}} \subseteq \mathcal{L}_2^0(\mathbb{P})$ (recall we defined this as the set of $\mathcal{L}_2(\mathbb{P})$ functions with $\mathbb{P}$ -mean zero). Some models are so “large”, however, that the inclusion is actually an equality. If the model is big enough (locally), you can move in any direction.

Example 3.2. Let $\mathcal{p}$ be the set of all distributions of continuous random variables (i.e. distributions with standard densities) and pick some point $\mathbb{P} \in \mathcal{P}$. For any bounded function $s \in \mathcal{L}_2^0(\mathbb{P})$ we can construct the convex path $p_\epsilon = (1 + \epsilon s)p_0$ (bounding s is required so that p_ϵ remains a density and the mean-square and pointwise scores coincide). As we saw in Example ??, s is the score for such a path at $\mathbb{P}$ and thus the tangent set at that point includes at least all bounded functions $s \in \mathcal{L}_2^0(\mathbb{P})$.

This is good enough for intuition, but to remove the boundedness condition on the scores and formally extend to all $\mathcal{L}_2^0(\mathbb{P})$ we can instead use the paths $p_\epsilon = \frac{e^{\lambda(\epsilon s)}}{\int e^{\lambda(\epsilon s(z))} p_0(z) dz} p_0$ with $\lambda(t) = 2(1 + e^{-2t})^{-1}$. These paths also have scores s and generate valid densities for any $s \in \mathcal{L}_2^0(\mathbb{P})$. For details see Example 1 in Chapter 3 of Bickel et al. (1993).

We give this kind of model its own name.

Locally Nonparametric Model

Definition 3.2. A model $\mathcal{P}$ is *locally nonparametric* at $\mathbb{P} \in \mathcal{P}$ if $\dot{\mathcal{P}}_{\mathbb{P}} = \mathcal{L}_2^0(\mathbb{P})$.

This is the kind of model we will work with most often for the remainder of this book. Locally nonparametric models are popular in practice because they enforce essentially no assumptions on the data-generating distribution. In fields like medicine, economics, or business, we often cannot say much about the world with great confidence (the distribution of variables, the functional form of their relationships, etc.). We will see shortly that the larger the model is, usually the more uncertainty we get in the final estimate. This makes sense: the more assumptions you’re willing to make, the more confident you can be of an estimate. But if you use a model that is too small (i.e. assume things that are not actually true) then the estimate may be biased and confidence intervals will typically underestimate the true sampling uncertainty. Working with a locally nonparametric model is therefore a maximally robust approach, or at worst conservative.

3.2.1 Factoring

There are, of course, many practical situations where we *do* know something. For example, in our familiar covariate-treatment-outcome (X, A, Y) setup, if the treatment is allocated randomly by coin flip, then we know for a fact that $\mathbb{P}(A = 1|X) = 1/2$. This is an important restriction on the model. Due to the ability to experimentally control treatment assignments, there are many cases in causal inference where one or more factors of a probability distribution are known exactly, but nothing is certain about the others. Thankfully, it's not hard to figure out the tangent set in such models.

To simplify things, we'll work in a generic model defined over two variables (Y, Z) (you can think of $Z = (A, X)$ if you please). We'll also presume again that all our distributions have densities and that the pointwise definition of the scores coincides with the mean-square derivative definition to facilitate the explanation. All of the results below also hold in the mean-square sense of scores in models without densities but everything is more cumbersome to prove without adding any intuition. I've put sketches of those proofs into boxed-off sections that you can safely ignore.

Since every distribution $\mathbb{P}(Y, Z)$ can be factored into $\mathbb{P}(Y, Z) = \mathbb{P}(Y|Z)\mathbb{P}(Z)$, every differentiable path of distributions $\mathbb{P}^{(\epsilon)}(Y, Z)$ induces a path of marginal distributions $\mathbb{P}^{(\epsilon)}(Z)$ and a path of conditional distributions $\mathbb{P}^{(\epsilon)}(Y|Z)$. Let's pick some arbitrary path with score s . We can take derivatives of the log-likelihoods along the induced conditional and marginal paths to obtain scores (in terms of the pointwise definition):

$$\begin{aligned}s_Z(z) &= \partial_0 \log p_\epsilon(z) \\ s_{Y|Z}(y, z) &= \partial_0 \log p_\epsilon(y|z)\end{aligned}$$

where I'm abbreviating the operator $\frac{d}{d\epsilon}\big|_{\epsilon=0}(\cdot)$ with $\partial_0(\cdot)$. When the meaning is clear from context I'll abuse notation a bit and write $s(z)$ for $s_Z(z)$ and similarly $s(y|z)$ for $s_{Y|Z}(y, z)$, the same way as is usually done for marginal and conditional density functions. If we iterate this over all differentiable paths and collect the resulting scores we generate what I'll call the marginal and conditional tangent sets $\dot{\mathcal{P}}_{\mathbb{P}}(Z)$ and $\dot{\mathcal{P}}_{\mathbb{P}}(Y|Z)$.

Let's see how these marginal and conditional scores are related to the joint score for a given path. First off, because

$$\log p_\epsilon(z, y) = \log(p_\epsilon(y|z)p_\epsilon(z)) = \log p_\epsilon(y|z) + \log p_\epsilon(z),$$

when we apply ∂_0 to both sides we get

$$s(y, z) = s(y|z) + s(z)$$

so the joint score is simply the sum of the marginal and conditional scores. But we can further characterize these scores in terms of the joint score. By the chain rule, $\partial \log p_\epsilon = \frac{\partial p_\epsilon}{p_\epsilon}$, so evaluating at $\epsilon = 0$ and shifting terms around gives

$$\partial_0 p_\epsilon = p_0 \partial_0 p_\epsilon = p_0 s. \quad (3.1)$$

This is true for the joint, conditional, and marginal densities/scores. If we start with this equation in terms of the joint density/score and integrate with respect to y on both sides we have

$$\int \partial_0 p_\epsilon(y, z) dy = \int p_0(y, z) s(y, z) dy.$$

On the left-hand-side, if we are allowed to exchange integration w.r.t z and differentiation w.r.t. ϵ (formally this is possible under mild restrictions on the model), then we get $\partial_0 \int p_\epsilon(y, z) dy = \partial_0 p_\epsilon(z) = p_0(z) s(z)$ by applying Equation ?? to the marginal $p_\epsilon(z)$. Dividing both sides of the above by $p_0(z)$ gives

$$s(z) = \int \frac{p_0(y, z)}{p_0(z)} s(y, z) dy = \int p(y|z) s(y, z) dy = \mathbb{E}[s(Y, Z)|Z = z]$$

and therefore $s(y|z) = s(y, z) - \mathbb{E}[s(Y, Z)|Z = z]$ since the conditional and marginal scores sum to the joint.

We now have explicit formulations of the marginal and conditional scores in terms of the joint score. Besides their relation to the joint score, these scores also have some interesting properties. You may recall from Definition ?? that $\mathbb{P}s = 0$, i.e. $\mathbb{E}[s(Y, Z)] = 0$. The conditional and joint score obey similar conditions:

$$\begin{aligned} \mathbb{E}[s(Z)] &= 0 \\ \mathbb{E}[s(Y|Z)|Z] &= 0. \end{aligned}$$

The first identity comes from the explicit definition of $s(z)$ because $\mathbb{E}_Z[s(Z)] = \mathbb{E}_Z[\mathbb{E}_Y[s(Y, Z)|Z]] = \mathbb{E}[s(Y, Z)] = 0$ by total expectation and since the joint score is mean zero. The second comes from

$$\begin{aligned} \mathbb{E}_Y[s(Y|Z)|Z] &= \mathbb{E}_Y[s(Y, Z) - \mathbb{E}_Y[s(Y, Z)|Z]|Z] \\ &= \mathbb{E}_Y[s(Y, Z)|Z] - \mathbb{E}_Y[s(Y, Z)|Z] \end{aligned}$$

We can also show that the marginal and conditional score functions are orthogonal to each other in $\mathcal{L}_2(\mathbb{P})$. Recall that two functions are orthogonal if their inner product is zero and that

the $\mathcal{L}_2(\mathbb{P})$ inner product of two functions is the expectation of their product (see Appendix ??). The expected value of the product of the conditional and marginal scores is

$$\mathbb{E}[s(Z)s(Y|Z)] = \mathbb{E}[\mathbb{E}[s(Y, Z)|Z](s(Y, Z) - \mathbb{E}[s(Y, Z)|Z])].$$

Abbreviating $S = s(Y, Z)$, by total expectation, $\mathbb{E}[\mathbb{E}_S[S|Z]S] = \mathbb{E}_Z[\mathbb{E}_S[\mathbb{E}_S[S|Z]S|Z]] = \mathbb{E}_Z[\mathbb{E}_S[S|Z]\mathbb{E}_S[S|Z]]$ so the above is zero.

This is actually a general fact about conditional expectation that is used in many places throughout probability theory: a random variable minus its conditional expectation is orthogonal to that conditional expectation. In fact, the conditional expectation $\mathbb{E}[f(Y, Z)|Z]$ of a random variable $f(Y, Z)$ is precisely its orthogonal projection onto the subspace spanned by all $\mathcal{L}_2(\mathbb{P})$ functions of Z alone: the best possible $\mathcal{L}_2(\mathbb{P})$ approximation of $f(y, z)$ that can be attained with functions of z . This is illustrated in Figure ??.

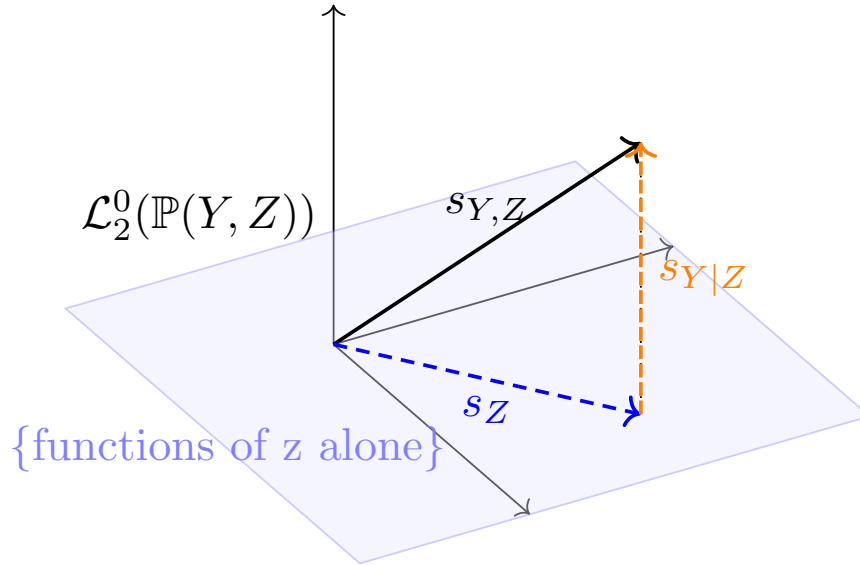


Figure 3.3: The marginal score is an orthogonal projection of the joint score and the conditional score is the residual.

Let's summarize all these results.

Conditional and Marginal Scores

Theorem 3.3. *Let $\mathcal{P} = \mathcal{P}(Y, Z)$ be a model over joint distributions $\mathbb{P}(Y, Z)$ of variables (Y, Z) and let $\mathcal{P}_\epsilon(Y, Z)$ be a differentiable path with score $s(y, z)$. This path induces marginal and conditional paths $\mathcal{P}_\epsilon(Z)$ and $\mathcal{P}_\epsilon(Y|Z)$ with scores*

$$s(z) = \mathbb{E}[s(Y, Z)|Z]$$

$$s(y|z) = s(y, z) - \mathbb{E}[s(Y, Z)|Z]$$

which sum to $s(y, z)$. The marginal score s_Z is the orthogonal projection of the joint score $s_{Y, Z}$ onto the span of all $\mathcal{L}_2(\mathbb{P})$ functions of z alone, while the conditional score $s_{Y|Z}$ is the residual of this projection. Therefore $s_{Y|Z} \perp s_Z$ in $\mathcal{L}_2(\mathbb{P})$. Moreover

$$\mathbb{E}[s(Z)] = 0$$

$$\mathbb{E}[s(Y|Z)|Z] = 0,$$

Proof Sketch for Mean-Square Scores

Let $\mathcal{P}_\epsilon$ be a (mean-square) differentiable path over distributions $\mathbb{P}(Y, Z)$ and let $\epsilon_n \rightarrow 0$ and $\mathbb{P}_n = \mathbb{P}_{\epsilon_n}$ be a sequence along the path with $\mathbb{P} = \mathbb{P}_{\epsilon=0}$. Let $\mathbb{Q}(Y, Z)$ be a dominating measure constructed for the sequence $\mathbb{P}_n$ and let $p_n(y, z) = \frac{d\mathbb{P}_n(Y, Z)}{d\mathbb{Q}(Y, Z)}$ be densities. Let $Q(Z)$ and $\mathbb{P}(Z)$ be the marginal distributions of $\mathbb{Q}(Y, Z)$ and $\mathbb{P}(Y, Z)$, respectively, so that $p_n(z) = \frac{d\mathbb{P}_n(Z)}{d\mathbb{Q}(Z)}$ are marginal densities.

Since the path of joint distributions is differentiable, there is a function $g(y, z)$ such that

$$\left\| \frac{\sqrt{p_n(y, z)} - \sqrt{p_0(y, z)}}{\epsilon_n} - g(y, z) \right\|_{\mathcal{L}_2(\mathbb{Q}(Y, Z))} \rightarrow 0$$

Our goal is to show that the path of marginal distributions is also differentiable, i.e. that there is a function $g(z)$ for which

$$\left\| \frac{\sqrt{p_n(z)} - \sqrt{p(z)}}{\epsilon_n} - g(z) \right\|_{\mathcal{L}_2(\mathbb{Q}(Z))} \rightarrow 0$$

First decompose

$$\sqrt{p_n(z)} - \sqrt{p(z)} = \frac{p_n(z) - p(z)}{\sqrt{p_n(z)} + \sqrt{p(z)}}$$

and notice that, by definition,

$$p_n(z) - p(z) = \int p_n(y, z) - p(y, z) d\mathbb{Q}(Y|Z)$$

$$= \int (\sqrt{p_n(y, z)} - \sqrt{p(y, z)}) (\sqrt{p_n(y, z)} + \sqrt{p(y, z)}) d\mathbb{Q}(Y|Z).$$

This is the equivalent of the more familiar formula $p(z) = \int p(y, z) dy$ except that in this case we do not assume that $\mathbb{Q}$ is a product measure. In the familiar case we see $dy = d\mu(Y)$ because $d\mu(Y) = d\mu(Y|Z)$, i.e. Lebesgue measure is a product measure.

Combining these observations,

$$\frac{\sqrt{p_n(z)} - \sqrt{p(z)}}{\epsilon_n} = \int \underbrace{\frac{\sqrt{p_n(y,z)} - \sqrt{p(y,z)}}{\epsilon_n}}_{g_n(y,z)} \underbrace{\frac{\sqrt{p_n(y,z)} + \sqrt{p(y,z)}}{\sqrt{p_n(z)} + \sqrt{p(z)}}}_{h_n(y,z)} dQ(Y|Z).$$

$g_n \xrightarrow{\mathcal{L}_2(Q(Y,Z))} g$ by assumption and since $\sqrt{p_n(y,z)} \xrightarrow{\mathcal{L}_2(Q(Y,Z))} \sqrt{p(y,z)}$, the second factor h_n can be shown to converge in $\mathcal{L}_2(Q(Y,Z))$ to $h(y,z) = \sqrt{p(y,z)/p(z)} = \sqrt{p(y|z)}$ with careful handling of the denominator.

Therefore one might conjecture that the $\mathcal{L}_2(Q(Z))$ limit of the above is $\int g(y,z)h(y,z) dQ(Y|Z)$ and indeed this is the case because

$$\begin{aligned} \int \left(\int g_n h_n - gh dQ(Y|Z) \right)^2 dQ(Z) &\leq \int \int (g_n h_n - gh)^2 dQ(Y|Z) dQ(Z) \\ &= \int (g_n h_n - gh)^2 dQ(Y,Z) \rightarrow 0 \end{aligned}$$

by Jensen's inequality ($0 \leq \mathbb{V}[U] = \mathbb{E}[U^2] - \mathbb{E}[U]^2 \implies \mathbb{E}[U]^2 < \mathbb{E}[U^2]$) and the aforementioned convergences.

This shows that $g(z) = \int g(y,z) \sqrt{p(y|z)} dQ(Y|Z)$. Defining $s(z) = 2g(z)/\sqrt{p(z)}$ and $s(y,z) = 2g(y,z)/\sqrt{p(y,z)}$ gives

$$s(z) = \int s(y,z) p(y|z) dQ(Y|Z) = \int s(y,z) dP(Y|Z) = \mathbb{E}[s(Y,Z)|Z=z].$$

Similar arguments may be used to derive the conditional score $s(y|z)$ and then the other facts follow algebraically.

Given this characterization, it's easy to derive the tangent set of a model that has a restricted factor. In a model with a restricted factor, paths can only change the other factors and the scores are the “directions” that change only the non-restricted factors. For example if $p(z)$ is fixed in the model but $p(y|z)$ is free, then the scores are $s(y,z) = s(y|z) + 0$, i.e. the z -component $s(z) = 0$ is fixed at 0 so as to not change the fixed marginal density $p(z)$.

Tangent Sets in Factored Models

Theorem 3.4. *Let $\mathcal{P}$ be locally nonparametric around some distribution $\mathbb{P}(Y,Z) = \mathbb{P}(Y|Z)\mathbb{P}(Z)$.*

Let $\mathcal{P}^{(Z)} = \{\mathbb{P}(Y|Z)\tilde{\mathbb{P}}(Z) : \tilde{\mathbb{P}} \in \mathcal{P}\}$, be a restriction of the original model where all the conditional distributions $\mathbb{P}(Y|Z)$ are replaced with the fixed conditional distribution $\mathbb{P}(Y|Z)$, but the marginal distributions of Z are allowed to vary. Then the tangent set at $\mathbb{P}$ in the restricted model is the set of all Z -marginal scores in the original model, i.e.

$$\dot{\mathcal{P}}_{\mathbb{P}}^{(Z)} = \dot{\mathcal{P}}_{\mathbb{P}}(Z),$$

which is all $\mathcal{L}_2^0(\mathbb{P})$ functions of z alone.

Similarly, if $\mathcal{P}^{(Y|Z)} = \{\tilde{\mathbb{P}}(Y|Z)\mathbb{P}(Z) : \tilde{\mathbb{P}} \in \mathcal{P}\}$, a mutation of the original model where all the marginal distributions are fixed, but the conditional distributions can still vary, then

$$\dot{\mathcal{P}}_{\mathbb{P}}^{(Y|Z)} = \dot{\mathcal{P}}_{\mathbb{P}}(Y|Z),$$

which is all $\mathcal{L}_2^0(\mathbb{P})$ functions of (y, z) with z -conditional mean zero.

3.3 Pathwise Differentiability

Now that we understand a bit more about the set of all scores we can return to the central identity above (definition Theorem ??). By comparing the asymptotic behavior while moving quickly along differentiable paths implied by regularity to the behavior implied by asymptotic linearity, we deduced that any RAL estimator (with influence function ϕ) satisfies

$$\dot{\Psi}_{\mathbb{P}}s = \mathbb{P}s\phi$$

for any differentiable path through $\mathbb{P}$ with score s . In other words, that equality is satisfied for all scores $s \in \dot{\mathcal{P}}_{\mathbb{P}}$, the tangent set of the model $\mathcal{P}$ at the distribution $\mathbb{P}$.

Let's dig into the left-hand term:

$$\dot{\Psi}_{\mathbb{P}}(s) \stackrel{\text{def}}{=} \lim_{\epsilon \rightarrow 0} \frac{\Psi(\mathbb{P}^{(\epsilon)}) - \Psi(\mathbb{P}^{(0)})}{\epsilon} = \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \Psi(\mathbb{P}^{(\epsilon)}).$$

For any fixed path this is a standard, vanilla derivative of a one-dimensional function of ϵ . It measures the (instantaneous) rate of change of the estimand as we shift the underlying distribution along the path away from $\epsilon = 0$. This is illustrated in Figure ??.

However, that slope is generally different depending on the path one chooses. There may be paths along with the estimand is increasing rapidly, paths along which it is decreasing, and paths along which it is constant. Moreover if you “accelerate” the speed at which a path goes by multiplying the score by some value $c > 1$, the derivative will also be magnified. That is why we write $\dot{\Psi}_{\mathbb{P}}(s)$ as a function of s , the “direction” (and “speed”) of the path. Since the slope is taken at $\epsilon = 0$, the only thing that matters is the immediate direction of the path at $\mathbb{P}$, i.e. the score s . The instantaneous rate of change of the estimand along two different paths with the same score will be the same. Because it is a derivative along a path, we aptly call the function $s \mapsto \dot{\Psi}_{\mathbb{P}}(s)$ mapping from $\dot{\mathcal{P}}_{\mathbb{P}}$ to $\mathbb{R}$ the *pathwise derivative* of the estimand Ψ at the distribution $\mathbb{P}$ in the model $\mathcal{P}$.

The identity $\dot{\Psi}_{\mathbb{P}}(s) = \mathbb{P}s\phi$ says that the slope $\dot{\Psi}_{\mathbb{P}}(s)$ may be found by taking the inner product of s with *any* influence function ϕ of a RAL estimator. In fact, it doesn't matter *which* RAL

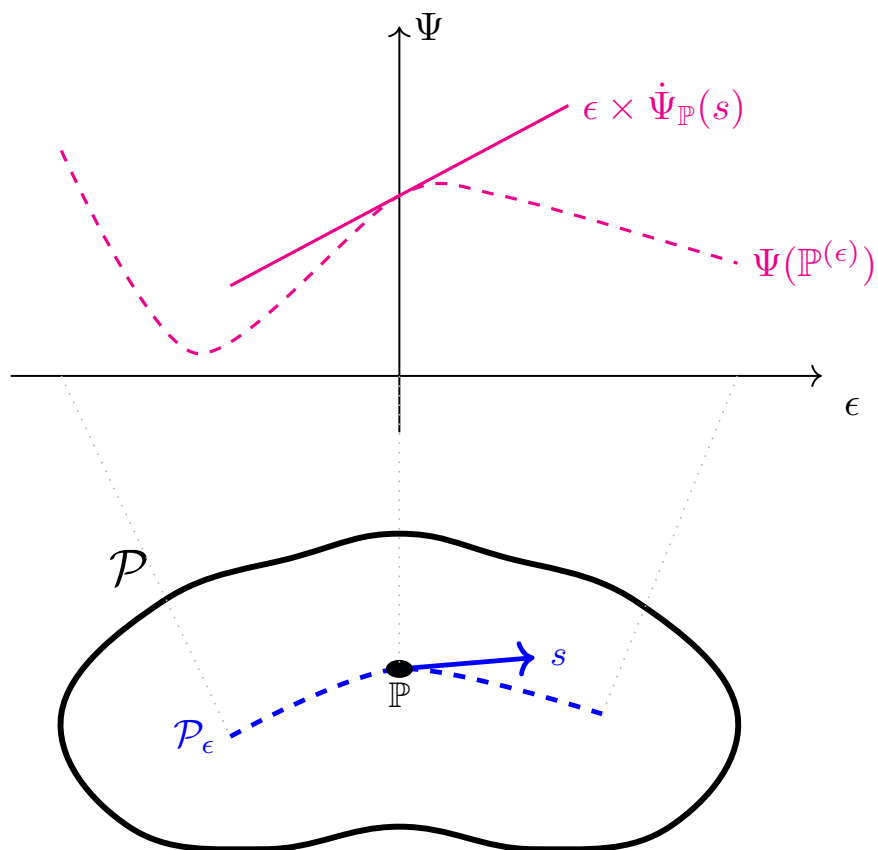


Figure 3.4: The pathwise derivative evaluated at a particular score s gives the rate of change of the estimand if we move the underlying distribution in the direction of (and at the speed of) the score.

estimator, if there are multiple: the result must be the same. We'll return to this point in the next section but we are actually getting ahead of ourselves because we are assuming there *is* a RAL estimator. The argument for Theorem ?? only established a condition that must be met by any RAL estimator, not that such an estimator exists. In fact, the result implies something about when RAL estimators *can* exist.

Notice that on the right-hand side, $\mathbb{P}s\phi$ is an inner product. Seen as a function $s \mapsto \mathbb{P}s\phi$ acting on the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$, the inner product is a bounded (i.e. continuous) linear functional (see Definition ?? and Example ??). Informally, a small change in the input score must always translate to a small change in the inner product with the fixed function ϕ . Since this inner product is equal to the pathwise derivative, that implies that the pathwise derivative must *also* be linear and continuous in the same sense: if we look at two paths that go in similar directions, the rate of change of the estimand must be similar on those paths.

The interesting thing is that the pathwise derivative has nothing to do with the estimator: it is defined purely by the model $\mathcal{P}$, the distribution $\mathbb{P}$ of interest within it, and the estimand Ψ . The pathwise derivative either exists and is linear and continuous across scores or it isn't. If it isn't a continuous linear functional, then it's not possible for it to be equal to any inner product. But since that equality must hold for any RAL estimator, if the pathwise derivative is not a continuous linear functional, then there cannot be any RAL estimator for that estimand in that model in the first place.

Pathwise Differentiability

Definition 3.3. An estimand Ψ is *pathwise differentiable* at $\mathbb{P} \in \mathcal{P}$ if the pathwise derivative $\dot{\Psi}_{\mathbb{P}}$ exists and is linear and continuous across the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$. If an estimand is not pathwise differentiable, there is no asymptotically linear estimator that is regular in that model.

Warning

Review Definition ?? for the definition of continuity and Section ?? for details and definitions relating to Hilbert spaces and Riesz representation. Understanding the details is not nearly as important as understanding the big-picture implications here so it's ok if you skip the mathematical details for now and come back to them on a second pass later.

It's sometimes helpful to think of the pathwise derivative as the best continuous, linear approximation to $\Psi(\mathbb{P})$ at the point $\mathbb{P}$, the same way that a tangent plane locally approximates the surface of a real-valued function of two or more variables. In this analogy, the tangent plane should itself be thought of as a function (a continuous and linear one) defined on the same variables.

Note

The analogy is somewhat imprecise because for a standard derivative of a function $f(z)$ with vector inputs, the “tangent plane” $\dot{f}(h) \approx f(z+h) - f(z)$ is defined on the same input vector space as the function itself, i.e. h and z are elements of the same vector space. For the pathwise derivative, the equivalent approximation is $\dot{\Psi}(s) \approx \Psi(\mathbb{P}^{(\epsilon)}) - \Psi(\mathbb{P})$, but $\mathbb{P}$ and s are two different kinds of objects. The trouble is that distributions in a model don’t naturally form a vector space: we can embed them into $\mathcal{L}_2$ as root-densities by assuming they are dominated by some other measure but at a loss of generality.

If that is done, however, then the pathwise derivative corresponds exactly with something called a Hadamard derivative. This kind of derivative is slightly weaker than the standard (total) derivative (called a Fréchet derivative). Roughly speaking, the usual total derivative tangent plane must approximate the error $\dot{f}(h) \approx f(z+h)$ for any sequence h approaching in any directions to 0, whereas the Hadamard derivative only needs to give the right approximation along sequences h that eventually settle into a single directional approach to zero. Roughly speaking, this corresponds to the pathwise derivative because the latter is only concerned with sequences along differentiable paths, i.e. paths where the difference of subsequent steps is eventually well-approximated by the “direction” function s (the score).

It is not necessary, however, to make assumptions in order to embed the model into $\mathcal{L}_2$. The pathwise derivative as given in Theorem ?? and Definition ?? works just fine in generic models. While most similar to the Hadamard derivative, it does not precisely correspond to any form of derivative that is defined on vector spaces.

Exercise 3.1. On a piece of paper sketch out a replica of Figure ??.

- Draw some curvy, smooth level sets of your choosing over the surface of $\sqrt{\mathcal{P}}$ representing the value of $\Psi(\mathbb{P})$ at each point. One of your level set lines should pass through the point $\sqrt{p}$.
- Draw some straight, evenly spaced level-set lines across $\sqrt{p}\dot{\mathcal{P}}$ representing the value of $\dot{\Psi}(s)$ at each point (you can peek at Figure ?? to see what I mean). One of the level-set lines should pass through $\sqrt{p}$ and at that point should be tangent to the level set of Ψ passing through $\sqrt{p}$ on $\sqrt{\mathcal{P}}$.

Your drawing represents the sense in which $\dot{\Psi}$ is a local, linear approximation to $\Psi : \mathcal{P} \rightarrow \mathbb{R}$. If you imagine zooming into the point $\sqrt{p} \in \sqrt{\mathcal{P}}$ the surface of the sphere begins to look essentially flat (like the tangent set) and the curves of the contours on the model also flatten out into a regular linear pattern. The flattening of the model is due to the differentiability of the paths whereas the flattening of the level sets are due to pathwise differentiability of the estimand.

A pathwise-differentiable estimand “varies smoothly” over the model the same way that a ordi-

nary function with a continuous derivative has no kinks. At any point of a kink or discontinuity in the estimand, no estimator can be regular.

Exercise 3.2. Let Z be normally distributed with known variance 1 but unknown mean μ , i.e. $\mathcal{P} = \{N(\mu, 1) : \mu \in \mathbb{R}\}$. Presume we are interested in estimating the quantity $\Psi(\mathbb{P}) = 1(\mathbb{E}_{\mathbb{P}}[Z] > 0)$, i.e. whether the mean is positive or not (we are estimating, not testing).

- Draw a plot of the function $\mu \mapsto \Psi(N(\mu, 1))$
- At $\mathbb{P} = N(0, 1)$, do you think Ψ is pathwise differentiable? If not, is it because the derivative doesn't exist or is it because the derivative isn't continuous?
- Repeat the above for the new estimand $\Psi(\mathbb{P}) = |\mathbb{E}_{\mathbb{P}}[Z]|$.
- Let $\hat{\mu} = \frac{1}{n} \sum Z_i$. What are the sampling distributions of $1(\hat{\mu} > 0)$ and $|\hat{\mu}|$ when $Z \sim N(0, 1)$?

This exercise shows that some simple estimands are not pathwise differentiable, although the examples are somewhat contrived because in both cases it would be more natural to estimate μ and then perform a hypothesis test. On the other hand, it also shows that lacking pathwise differentiability need not be the end of the world: both of the estimands have natural, easily-understood estimators that would be hard to beat in practice even in nonparametric models.

The examples in Exercise ?? are contrived. More often the types of “kinks” or discontinuities that cause problems are difficult to visualize in lower-dimensions. It's not as if the value of most practically interesting non-differentiable estimands changes suddenly as we move along a single differentiable path. The fact that the path is differentiable means that the densities change smoothly as we walk in that fixed direction and that usually implies smooth behavior of the estimand along a given path.

The kinks are sometimes only visible “rotationally”. I like to imagine standing at a point $\mathbb{P}$ on the “surface” $\Psi(\mathbb{P})$ and looking in some direction: I imagine seeing how steep the ascent or descent is in whatever direction I look. The problem with most practically interesting estimands that are not differentiable isn't that there is a discontinuity or kink in any given direction, it's that no matter what direction you look at, there is always another direction where the slope is steeper. But an estimand is pathwise differentiable at a point in a model only if there is a “direction of maximum slope” away from that point. In infinite dimensions, kinks occur in this “rotational” sense more often than they occur in the more familiar one-dimension sense.

Example 3.3. Let $\mathcal{P}$ be all distributions of a continuous variable Z . Let's say that our goal is to estimate the value of the density function of z at the point $z = 0$. There generally isn't a good motivation for wanting to know this, it's just a simple example that illustrates ideas that apply identically to more complicated estimands.

We'll show that there is no RAL estimator of the density at a point by establishing that the pathwise derivative is not bounded over the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$ (at any $\mathbb{P}$). The intuition is that we can find a path that comes very, very close to modifying the density *only* at the point $z = 0$. In other words, we can modify the estimand by arbitrary amounts essentially without changing the overall density.

Fix a distribution $\mathbb{P}$. Since the model is nonparametric, the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$ is all of $\mathcal{L}_2^0(\mathbb{P})$. Since that's the case, any zero-mean $\mathcal{L}_2(\mathbb{P})$ function can be a score.

Define the sequence of probabilities $t_n = \mathbb{P}\{-1/n \leq Z \leq 1/n\}$. Since these are probabilities of smaller and smaller sets over a continuous random variable they must shrink to zero.

We'll consider a sequence of scores

$$s_n(z) = \frac{1(-\frac{1}{n} \leq z \leq \frac{1}{n}) - t_n}{\sqrt{t_n(1-t_n)}}$$

The random variable $1(-1/n \leq Z \leq 1/n)$ is 1 with probability t_n and 0 with probability $1-t_n$, i.e. it is $\text{Bern}(t_n)$. Therefore it has mean t_n and standard deviation $\sqrt{t_n(1-t_n)}$. Consequently, the mean of $s_n(Z)$ is zero and its standard deviation ($\mathcal{L}_2(\mathbb{P})$ norm) is 1. Thus s_n are all $\mathcal{L}_2(\mathbb{P})$ functions with mean zero as desired, i.e. these are scores in the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$. Moreover these are unit-norm scores.

To find the pathwise derivatives $\dot{\Psi}(s_n)$, let's consider a sequence of paths $\mathcal{P}_\epsilon^{(n)}$ that have the desired scores. For this we can use the convex paths $p_\epsilon^{(n)} = (1 + \epsilon s_n)p$ that we previously used in Example ?? and Example ??. Now we can evaluate the pathwise derivative directly:

$$\begin{aligned} \dot{\Psi}(s_n) &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \Psi(\mathbb{P}^{(\epsilon)}) \\ &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} (1 + \epsilon s_n(0))p(0) \\ &= s_n(0)p(0) \\ &= \frac{1-t_n}{\sqrt{t_n(1-t_n)}}p(0) \end{aligned}$$

Since t_n converges to zero, the denominator blows up. Since we only considered unit scores, we proved that there is no pure direction of maximum increase: no matter what direction we look, there is always another direction for which the pathwise derivative is bigger. Therefore the pathwise derivative is unbounded (not continuous). As a consequence, there can be no RAL estimator for the density at a point when the model is all continuous densities.

With more setup and more care taken to define tangent sets, etc., very similar arguments apply if we are estimating the value of a prediction function at a point (i.e. $\mathbb{E}[Y|X = x]$) or a conditional average treatment effect ($\mathbb{E}[Y|X = x, A = 1] - \mathbb{E}[Y|X = x, A = 0]$) in

nonparametric models. These are important examples of practically relevant estimands for which there are no RAL estimators! The theory we develop in the rest of this book does not apply to these estimands which is why approaches to estimate them can be more ad-hoc, context dependent, and difficult to build formal inference around.

Exercise 3.3. Whether an estimand is pathwise differentiable or not depends on the model. Imagine that we know Z is normally distributed with known variance 1 but unknown mean μ , i.e. $\mathcal{P} = \{N(\mu, 1) : \mu \in \mathbb{R}\}$ and we are interested in estimating the value of the density $p(z)$ at the point $z = 0$. This is again a contrived example but the point is to illustrate ideas in the simplest setting possible. Recall that the density of a $N(\mu, 1)$ random variable is $p(z) = (2\pi)^{-1/2}e^{-(z-\mu)^2/2}$.

- What is the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$ at any point $\mathbb{P}$ in this model? What is the dimension of this tangent set?
- For any score $s \in \dot{\mathcal{P}}_{\mathbb{P}}$, construct the convex path $p_{\epsilon} = (1 + \epsilon s)p$ and use it to compute the pathwise derivative $\dot{\Psi}(s)$.
- What is $\dot{\Psi}(s)/\|s\|$? According to Definition ?? and Definition ?? what does this imply about the pathwise differentiability of the estimand in this model?
- Can you guess what a RAL estimator of $p(0)$ might be?

Pathwise differentiability does not necessarily imply that there *is* a RAL estimator. It only allows one to exist. In the next part of this book we will study how to actually build efficient estimators and we will see that we need certain assumptions even to guarantee asymptotic linearity. Whether or not that can be done and in what context depends on estimand in question.

3.4 Gradients

Tangent sets and pathwise derivatives are important tools that have already proven useful to us in characterizing when RAL estimators can even exist, but we can go further. In this section we'll build on these concepts to fully characterize all possible RAL estimators of any estimand in a pathwise differentiable model.

From here on out we will presume that the estimand is pathwise differentiable so that it is possible for the central identity $\dot{\Psi}_{\mathbb{P}}(s) = \mathbb{P}(s\phi)$ for all $s \in \dot{\mathcal{P}}_{\mathbb{P}}$ to hold for some function ϕ . Our goal is to characterize the set of functions ϕ that can be influence functions, i.e. the set of functions that satisfy this identity. Once we know what all the possible options are for influence functions, we can easily compare them and choose the most “efficient” influence function to build an estimator around.

As previously discussed, if a function satisfies the identity it does not immediately mean that a real-world RAL estimator can be constructed, but we will say that it implies that RAL

estimator exists “in principle”. For that reason, we’ll say that any function ϕ that satisfies the central identity is an influence function of a RAL estimator, even if such an estimator can’t be built in practice without additional assumptions. We will deal with understanding those additional assumptions later.

Warning

At this point, understanding Riesz representation becomes important. I encourage you to read Section ?? and any other background material in Appendix ?? you need before proceeding. I promise you will find the material rewarding and immediately useful.

All of this starts with pathwise differentiability. For a pathwise differentiable estimand, we know that $\dot{\Psi}$ is a bounded linear functional defined on the tangent set. In order to use an important result in functional analysis called the Riesz representation theorem we need to extend the pathwise derivative function $\dot{\Psi}$ so it is defined on an entire vector space. Although the tangent set is a vector space in all of the major examples we consider in this book, there are cases where it is not. I’ll use the symbol $\dot{\mathcal{P}}_{\mathbb{P}}$ to denote the linear closure of the tangent set, which I’ll call the tangent *space*.

Often $\dot{\mathcal{P}} = \dot{\mathcal{P}}$ already. A linear closure of a set of vectors is all possible linear combinations of those vectors along with any limits. Since it is linear and continuous, the pathwise derivative is easily defined on $\dot{\mathcal{P}}_{\mathbb{P}}$. Any score function in this space is either a) a linear combination $\bar{s} = \sum c_j s_j$ of scores $s_j \in \mathcal{P}_{\mathbb{P}}$, in which case we define $\dot{\Psi}(\bar{s}) = \sum c_j \dot{\Psi}(s)$ in accordance with linearity; or b) a limit of such combinations $\bar{s} = \lim_{n \rightarrow \infty} s_n$, in which case we define $\dot{\Psi}(\bar{s}) = \lim_{n \rightarrow \infty} \dot{\Psi}(s)$ in accordance with continuity. $\dot{\mathcal{P}}_{\mathbb{P}}$ is always a subspace of $\mathcal{L}_2(\mathbb{P})$ and thus inherits its inner product $\langle f, g \rangle = \int fg d\mathbb{P} = \mathbb{P}fg$.

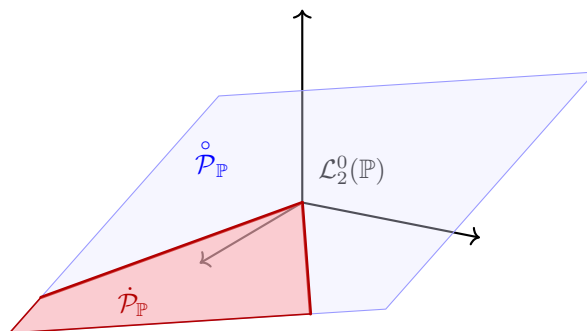


Figure 3.5: Cartoon of the relationship between the tangent set $\dot{\mathcal{P}}_{\mathbb{P}}$ and tangent space $\dot{\mathcal{P}}_{\mathbb{P}}$.

The tangent set is not always a subspace but it is always closed under positive scaling (prove it!), which is why it appears as a wedge in the cartoon. The tangent space fills in the rest of the plane that the tangent set sits in, i.e. it is the smallest subspace of $\mathcal{L}_2^0(\mathbb{P})$ that contains the tangent set.

Exercise 3.4. For two models $\mathcal{Q} \subset \mathcal{P}$, what is the relationship between $\dot{Q}_{\mathbb{P}}$ and $\dot{\mathcal{P}}_{\mathbb{P}}$ at any point $\mathbb{P} \in \mathcal{Q}$?

When the estimand is pathwise differentiable, the pathwise derivative $\dot{\Psi}$ is a bounded linear functional defined on the full tangent space $\dot{\mathcal{P}}_{\mathbb{P}}$. Since it is bounded and linear, $\dot{\Psi}$ has some direction of steepest ascent and moreover the Riesz representation theorem formally guarantees that the application of $\dot{\Psi}$ to an argument s is equivalent to taking the inner product of s with some unique element of $\dot{\mathcal{P}}_{\mathbb{P}}$ that points in that direction of steepest ascent. This is completely analogous to the way that the derivative in a direction $v \in \mathbb{R}^d$ of a multivariable function $f(z)$ at z_0 can be computed by taking the dot product of that direction vector with the *gradient* of f at z_0 :

$$\dot{f}_{z_0}(v) = \lim_{\epsilon \rightarrow 0} \frac{f(z_0 + \epsilon v) - f(z_0)}{\epsilon} = v \cdot \nabla f.$$

In $\mathbb{R}^d$, the gradient vector ∇f is precisely the Riesz representer of the directional derivative $\dot{f}$ (at z_0 , but this is suppressed in the notation). For that reason, we'll use $\nabla \Psi$ to denote the Riesz representer of the pathwise derivative $\dot{\Psi}_{\mathbb{P}}$ at $\mathbb{P} \in \mathcal{P}$ and call it the “gradient” of the estimand at that point. The direction of the Riesz representer corresponds to the direction of maximum slope for a linear functional. In the case of the pathwise derivative, this corresponds to the direction of maximum instantaneous rate of change of the estimand across paths with unit-norm scores.

Gradient of a Pathwise Differentiable Estimand

Definition 3.4. For an estimand Ψ that is pathwise differentiable at $\mathbb{P} \in \mathcal{P}$, the *gradient* (or *canonical gradient*) $\nabla_{\mathbb{P}} \Psi$ is the (unique) Riesz representer of the pathwise derivative $\dot{\Psi} : \dot{\mathcal{P}}_{\mathbb{P}} \rightarrow \mathbb{R}$.

The gradient is always an element of $\dot{\mathcal{P}}_{\mathbb{P}} \subseteq \mathcal{L}_2^0(\mathbb{P})$ and is thus square integrable and mean zero under $\mathbb{P}$ at a minimum.

The gradient represents the “direction” in which the estimand is increasing most quickly and its $\mathcal{L}_2(\mathbb{P})$ norm $\|\nabla_{\mathbb{P}} \Psi\| = \dot{\Psi}(\frac{\nabla_{\mathbb{P}} \Psi}{\|\nabla_{\mathbb{P}} \Psi\|})$ encodes the speed at which it is instantaneously increasing in that direction.

For functions defined on $\mathbb{R}^d$ the gradient is a familiar d -dimensional vector. The pathwise derivative is defined on the tangent space $\dot{\mathcal{P}}_{\mathbb{P}}$, a subspace of the space of functions $\mathcal{L}_2^0(\mathbb{P})$, and therefore the gradient of the estimand is also an $\mathcal{L}_2^0(\mathbb{P})$ function: it must be square integrable and have mean zero under $\mathbb{P}$. The gradient depends on what point $\mathbb{P}$ the pathwise derivative is taken at and in what model $\mathcal{P}$, but these dependencies are often suppressed in the notation.

The gradient is the Riesz representer of the pathwise derivative, so, by definition,

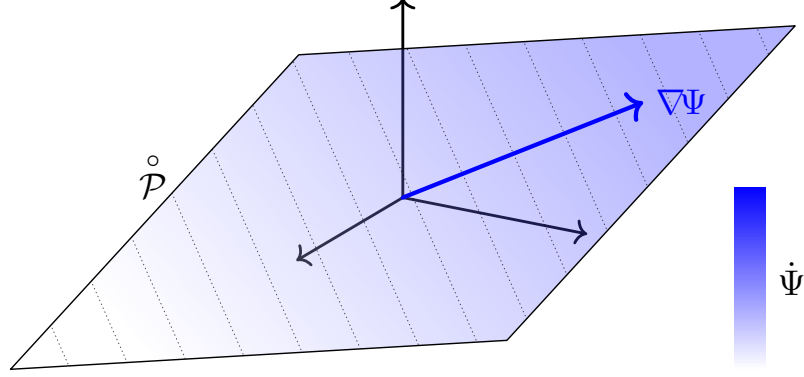


Figure 3.6: Visual representation of the gradient. The value of the pathwise derivative function $\dot{\Psi}_{\mathbb{P}}$ is depicted on the domain $\dot{\mathcal{P}}_{\mathbb{P}}$ using the blue shading: darker values indicate higher values of $\dot{\Psi}_{\mathbb{P}}(s)$ at each point $s \in \dot{\mathcal{P}}_{\mathbb{P}}$. Thin dotted lines represent the level sets of the pathwise derivative. The gradient $\nabla \Psi$ is shown as the dark blue arrow, representing the direction of fastest ascent of the linear function $\dot{\Psi}_{\mathbb{P}}$ and the speed of that ascent (spacing of the level sets; the closer together, the bigger the gradient). Since $\dot{\Psi}_{\mathbb{P}}$ is a derivative, larger values correspond to larger rates of increase of the estimand Ψ , meaning that the gradient corresponds to the direction of steepest instantaneous rate of increase of the estimand.

$$\dot{\Psi}_{\mathbb{P}}(s) = \mathbb{P}s \nabla \Psi \quad \text{for all } s \in \dot{\mathcal{P}}_{\mathbb{P}}.$$

This looks exactly like our central identity: $\dot{\Psi}_{\mathbb{P}}(s) = \mathbb{P}s \phi$ for all $s \in \dot{\mathcal{P}}_{\mathbb{P}}$. Since the gradient satisfies the identity for all scores in the tangent space, it certainly satisfies it for the subset of those that are in the tangent *set*. Therefore the gradient is an influence function for a RAL estimator! Since the gradient always exists (by Riesz representation), we have proven that every pathwise differentiable estimand has at least one RAL estimator. In fact, there may be many RAL estimators with this influence function, but by virtue of having the same influence function they are asymptotically equivalent.

Exercise 3.5. Use the identity above to prove the claim in Definition ?? that $\|\nabla_{\mathbb{P}} \Psi\| = \dot{\Psi}(\frac{\nabla_{\mathbb{P}} \Psi}{\|\nabla_{\mathbb{P}} \Psi\|})$.

Example 3.4. Let $\mathcal{P}$ be the set of all possible distributions of continuous random variables and presume the estimand of interest is the mean $\Psi(\mathbb{P}) = \mathbb{E}_{\mathbb{P}}[Z] = \int zp(z) dz$. Let $\psi_{\mathbb{P}}$ abbreviate the mean of Z for a distribution $\mathbb{P}$. We'll see that the gradient for this estimand is easy to find in this model moreover that we can even construct an estimator that has the gradient as its influence function.

We have already shown that this model is locally nonparametric (see Example ??) so at any $\mathbb{P}$ the tangent set is $\dot{\mathcal{P}}_{\mathbb{P}} = \mathcal{L}_2^0(\mathbb{P})$. For any bounded score $s \in \mathcal{L}_2^0(\mathbb{P})$ we can construct the usual convex path $p_{\epsilon} = (1 + \epsilon s)p$ which has score s . For unbounded scores we could construct the more complicated paths described in Example ?? but this adds complexity without adding understanding so we'll explain using the simpler convex paths.

For a convex path the pathwise derivative can be evaluated explicitly assuming enough regularity to pass the derivative through the integral:

$$\begin{aligned}\dot{\Psi}(s) &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \Psi((1 + \epsilon s)p) \\ &= \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \int z(1 + \epsilon s(z))p(z) dz \\ &= \int \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} z(1 + \epsilon s(z))p(z) dz \\ &= \int z s(z) p(z) dz\end{aligned}$$

This last expression is exactly $\langle z \mapsto z, s \rangle_{\mathcal{L}_2(\mathbb{P})}$, i.e. an inner product with the score. We might be tempted to say that the gradient is $f(z) = z$ but this isn't right. The Riesz representer must be a function in the domain of the linear functional, which in this case is $\mathcal{L}_2^0(\mathbb{P})$, zero-mean functions. But $f(z) = z$ is not zero-mean under $\mathbb{P}$, i.e. $\mathbb{E}_{\mathbb{P}}[f(Z)] = \psi_{\mathbb{P}} \neq 0$ in general, so this cannot be the gradient.

However, the situation is easily remedied. Any score s is also mean-zero under $\mathbb{P}$, so

$$0 = \int s(z)p(z) dz = \int \psi_{\mathbb{P}} s(z)p(z) dz,$$

meaning that we can add this term to what we have above and obtain

$$\dot{\Psi}(s) = \int z s(z)p(z) dz + \underbrace{\int \psi_{\mathbb{P}} s(z)p(z) dz}_0 = \int (z - \psi_{\mathbb{P}}) s(z)p(z) dz.$$

The function $z \mapsto z - \psi_{\mathbb{P}}$ is mean zero under $\mathbb{P}$ so it is in the tangent space. The equality above establishes that this is the gradient of the mean.

We saw in Example ?? that $\phi(z) = z - \psi_{\mathbb{P}}$ is exactly the influence function of the sample mean $\hat{\psi} = \frac{1}{n} \sum Z_i$. We already know the sample mean is regular but the fact that its influence function satisfies Theorem ?? also proves that regularity.

Exercise 3.6. The gradient generally depends on the model, even though we hide this dependence in the notation. Fix the estimand Ψ and a distribution $\mathbb{P} \in \mathcal{Q} \subset \mathcal{P}$. Let $\nabla_{\mathbb{P} \in \mathcal{Q}} \Psi$ be the gradient in the smaller model and let $\nabla_{\mathbb{P} \in \mathcal{P}} \Psi$ be the gradient in the larger model. Argue for why $\|\nabla_{\mathbb{P} \in \mathcal{Q}} \Psi\| \leq \|\nabla_{\mathbb{P} \in \mathcal{P}} \Psi\|$. *Hint:* what is the relationship between the tangent spaces?

3.4.1 Other Influence Functions

Using Riesz representation, we showed that for any pathwise differentiable estimand there is at least one function that can be the influence function of a RAL estimator (in principle). Now we will understand in what cases there can be RAL estimators with influence functions different than the gradient.

In general, Riesz representation says that the gradient is the *only* element ϕ in the tangent space for which $\dot{\Psi}(s) = \mathbb{P}s\phi$ for all scores in the tangent space. When the model is locally nonparametric at $\mathbb{P}$, that means the gradient is the only mean-zero, square-integrable function that satisfies that identity. But since any influence function must also be mean-zero and square integrable (Definition ??), that rules out the possibility that any other $\mathcal{L}_2^0(\mathbb{P})$ function can be an influence function. Therefore in locally nonparametric models there is only *one* influence function (the gradient of the estimand) and thus all RAL estimators are asymptotically equivalent to each other.

By itself this fact is already incredibly useful since locally nonparametric models are among the most practical in the real world. We'll see in the next chapter that deriving the gradient (and thus the only influence function) in a nonparametric model is easy to do.

Things are more interesting when the model is not locally nonparametric, i.e. when the tangent space is *not* all of $\mathcal{L}_2^0(\mathbb{P})$. Since the tangent space is a vector space, it has an orthogonal complement $\dot{\mathcal{P}}_{\mathbb{P}}^{\perp}$ relative to $\mathcal{L}_2^0(\mathbb{P})$. This is essentially the largest subspace of directions in which we are not allowed to perturb the distribution $\mathbb{P}$ because doing so would take us out of the model. The elements of this space therefore cannot be scores at $\mathbb{P} \in \mathcal{P}$ but nonetheless they live with scores in the common space $\mathcal{L}_2^0(\mathbb{P})$.

Warning

If need be, you should review Section ?? and specifically Definition ?? to learn more about orthogonal complements in Hilbert spaces.

By definition, any element $v \in \dot{\mathcal{P}}^{\perp}$ is orthogonal to any score $s \in \dot{\mathcal{P}}$. Some elements of the tangent *space* may not actually be scores but we'll refer to them as such anyways. Therefore $0 = \langle s, v \rangle_{\mathcal{L}_2(\mathbb{P})} = \mathbb{P}sv$ for *any* s . Starting with our influence function identity with $\phi = \nabla\Psi$, we can therefore add $0 = \mathbb{P}sv$ to the right-hand side without violating the equality:

$$\dot{\Psi}_{\mathbb{P}}(s) = \mathbb{P}s(\nabla\Psi + v) \quad \text{for all } s \in \dot{\mathcal{P}}_{\mathbb{P}}.$$

Moreover the functions v in the orthogonal complement of the tangent space are the *only* functions for which this holds. This fact is formalized in Theorem ?? and illustrated graphically in Figure ??.

Set of Influence Functions

Theorem 3.5. *Any influence function of a RAL estimator of a pathwise differentiable estimand is an element of the set $\Phi_{\mathbb{P}} = \{\nabla_{\mathbb{P}}\Psi + v : v \in \dot{\mathcal{P}}_{\mathbb{P}}^{\perp}\}$ where $\dot{\mathcal{P}}_{\mathbb{P}}^{\perp}$ is the orthogonal complement of the tangent space $\dot{\mathcal{P}}_{\mathbb{P}}$ relative to $\mathcal{L}_2^0(\mathbb{P})$. Any element of the set Φ is, in principle, the influence function of some RAL estimator.*

In a locally nonparametric model, $\dot{\mathcal{P}}_{\mathbb{P}}^{\perp}$ only contains the zero function so $\nabla_{\mathbb{P}}\Psi$ is the only possible influence function of a RAL estimator.

As a corollary of Theorem ??, any estimator that is asymptotically linear at $\mathbb{P}$ with an influence function in this set must be regular at $\mathbb{P} \in \mathcal{P}$.

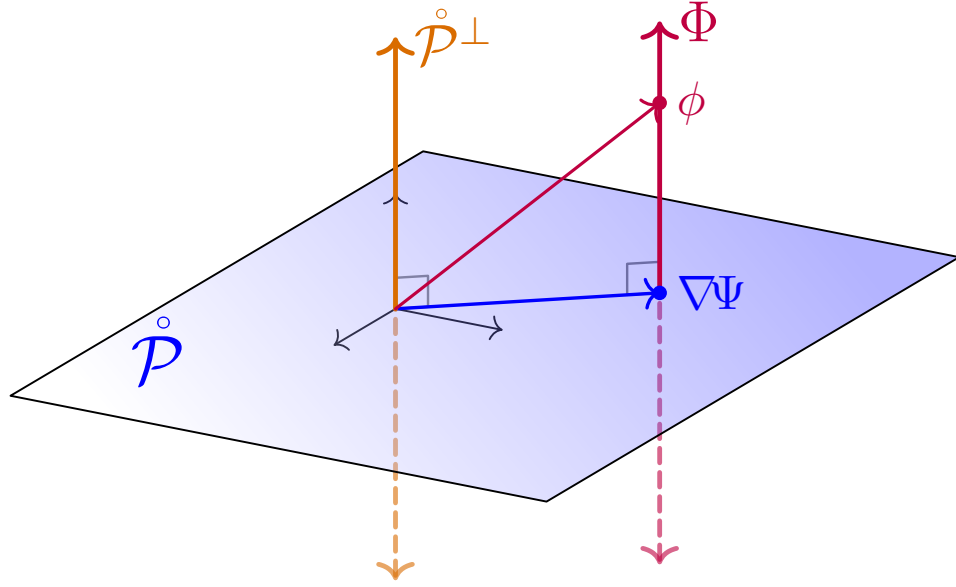


Figure 3.7: The set of influence functions Φ is the orthogonal complement of the tangent space $\dot{\mathcal{P}}^{\perp}$ translated over to the point $\nabla\Psi$ (a translated subspace is called a *linear variety*). The canonical gradient $\nabla\Psi$ has smaller norm (“length”) than any other influence function $\phi \in \Phi$.

From here on out we’ll call any element of the set $\{\nabla\Psi + v : v \in \dot{\mathcal{P}}^{\perp}\}$ a “gradient” of Ψ , reserving the term *canonical gradient* for $\nabla\Psi$. The term gradient is justified for any element $\phi \in \Phi$ because the defining identity $\dot{\Psi}(s) = \mathbb{P}s\phi$ holds for all $s \in \dot{\mathcal{P}}$. The only difference is that,

unlike the canonical gradient, the functions ϕ are not necessarily elements of tangent space, the domain of the linear functional $\dot{\Psi}$.

3.5 Efficiency

Theorem ?? is an extremely powerful result because it succeeds in completely characterizing the set of influence functions of RAL estimators. That was the goal we set out to achieve and now we can reap the reward.

The benefit of understanding what all the possible influence functions are is that we can now find the most “efficient” one. In Section ?? we saw that the central limit theorem immediately implies that any asymptotically linear estimator with influence function ϕ has asymptotic variance equal to $\mathbb{V}[\phi(Z)] = \|\phi\|^2$. Given a set of possible influence functions Φ , all we need to do to find the one corresponding to the most efficient RAL estimator is figure out which of them has smallest norm.

Theorem ?? makes this easy. Pick any influence function $\phi \in \Phi$. By Theorem ??, that function can be written as $\phi = \nabla\Psi + v$ for some $v \in \dot{\mathcal{P}}^\perp$. Moreover, since $\nabla\Psi$ is in the tangent space and v is some vector orthogonal to the tangent space, these two vectors create the sides of a right triangle with ϕ as the hypotenuse. Thus by the Pythagorean Theorem (Theorem ??),

$$\|\phi\|^2 = \|\nabla\Psi\|^2 + \|v\|^2 \geq \|\nabla\Psi\|^2.$$

In other words, the canonical gradient is the influence function with the smallest norm: the *efficient influence function*. This is visually obvious in Figure ??.

Efficient Influence Function

For a pathwise differentiable estimand Ψ in a given model $\mathcal{P}$, the influence function of the most efficient RAL estimator is equal to the canonical gradient $\nabla_{\mathbb{P}}\Psi$. We call this the efficient influence function and write it $\phi_{\mathbb{P}}^*$.

The efficient influence function and the canonical gradient are the same function, but they have two different definitions. The canonical gradient encodes the direction and magnitude of steepest increase of the estimand within the model and has nothing to do with any estimator. The efficient influence function encodes the asymptotic error of the most efficient RAL estimator and has nothing to do with any model. The fact that these two different objects are the same function is remarkable. Therefore whether we choose to call this function the canonical gradient or the efficient influence function is a matter of perspective. The former is more appropriate if we are discussing models, the latter if we are discussing estimators.

Because the canonical gradient depends on the estimand Ψ , the distribution in question $\mathbb{P}$, and the model it lives in $\mathcal{P}$, the efficient influence function (corresponding to the most efficient

RAL estimator) has the same dependencies. The first dependence is obvious: what the best estimator is obviously depends on what is being estimated. The second we have already discussed in Chapter ??: the influence function of any fixed asymptotically linear estimator depends on what distribution the data for that estimator are drawn from.

The third dependence is also natural. The “size” of the model corresponds to our existing knowledge of the world or the assumptions we’re willing to make. The more we know, the fewer distributions remain in the model. The smaller the model, the smaller the tangent space at any given point within it (it won’t be bigger, at a minimum). The smaller the tangent space, the larger the orthogonal complement, meaning we have more influence functions to choose from in order to find one with the smallest norm. The smaller the norm of the influence function, the smaller the asymptotic variance and the more certain our estimates are. This is the mechanism by which more a-priori knowledge about the world translates to more confident estimates, as one would expect.

Example 3.5. In Example ?? we showed that the canonical gradient of the mean estimand in the nonparametric model is $\nabla_{\mathbb{P}}\Psi(z) = z - \mathbb{E}_{\mathbb{P}}[Z]$ which has squared norm equal to $\mathbb{V}_{\mathbb{P}}[Z]$. If $Z \sim \mathbf{N}(0, 1)$, for example, the squared norm of the gradient is 1.

In this example we’ll describe how the canonical gradient changes and its norm shrinks when the model is further restricted. We’ll pick a somewhat contrived model to illustrate the point.

Let $\mathbb{P} = \mathbf{N}(0, 1)$ and consider the model of densities $\mathcal{P} = \{(1_{\epsilon}s)p : \epsilon \in \mathbb{R}\}$ where $s(z) = z - z^3$ which has mean zero. Obviously this is a one-dimensional model (a path with score s) so the canonical gradient at $\mathbb{P}$ must be a scalar multiple of s , i.e. $\nabla_{\mathbb{P}}\Psi = cs$. We can figure out what the constant c is because the squared norm of the canonical gradient must be equal to the pathwise derivative in the direction of the canonical gradient: $\dot{\Psi}(cs) = \mathbb{P}(cs)(cs)$. Therefore

$$c = \frac{\dot{\Psi}(s)}{\mathbb{P}s^2} = \frac{\mathbb{E}_{\mathbb{P}}[Z^2 - Z^4]}{\mathbb{E}_{\mathbb{P}}[Z^2 - 2Z^4 + Z^6]}$$

because we already showed in Example ?? that the pathwise derivative for the mean in this type of path is $\mathbb{E}_{\mathbb{P}}[Zs(Z)]$. Evaluating the above at $\mathbb{P} = \mathbf{N}(0, 1)$ gives

$$c = \frac{1 - 3}{1 - 2 \times 3 + 15} = -\frac{1}{5}$$

Therefore the squared norm of the canonical gradient is $\|\nabla\Psi\|^2 = c^2\|s\|^2 = \frac{2}{5}$.

This is about half the size of the norm of the canonical gradient for the same estimand in the nonparametric model. Since the canonical gradient in the nonparametric model corresponds to the influence function of the sample mean, the implication is that in this restricted model there is a RAL estimator of the population mean that is about twice as efficient as the sample mean estimator at this distribution!

The score I chose for the model in Example ?? was fairly arbitrary. In fact you can repeat the exercise with any score that is not a scalar multiple of $z - \mathbb{E}_{\mathbb{P}}[Z]$ and obtain a similar result. More generally you can choose any model such that its tangent space is smaller and no longer includes the original canonical gradient in the nonparametric model.

Exercise 3.7. Fix the estimand Ψ and a distribution $\mathbb{P} \in \mathcal{Q} \subset \mathcal{P}$. Let $\nabla_{\mathbb{P} \in \mathcal{Q}} \Psi$ be the canonical gradient in the smaller model and let $\nabla_{\mathbb{P} \in \mathcal{P}} \Psi$ be the canonical gradient in the larger model and presume that the tangent space for the smaller model does not include $\nabla_{\mathbb{P} \in \mathcal{P}} \Psi$. Prove that $\nabla_{\mathbb{P} \in \mathcal{P}} \Psi$ is still a gradient in the smaller model, i.e.

$$\dot{\Psi}_{\mathbb{P}}(s) = \mathbb{P}s \nabla_{\mathbb{P} \in \mathcal{P}} \Psi \quad \text{for all } s \in \dot{\mathcal{Q}}_{\mathbb{P}}.$$

Why does this imply that the gradient in the smaller model $\nabla_{\mathbb{P} \in \mathcal{Q}} \Psi$ is an orthogonal projection of $\nabla_{\mathbb{P} \in \mathcal{P}} \Psi$ onto the smaller tangent space $\dot{\mathcal{Q}}_{\mathbb{P}}$?

The relationship between the norm of the canonical gradient and the pathwise derivative means that the “speed limit” for asymptotic variance in estimation is directly related to how quickly the estimand can change under small perturbations of the data-generating distribution within the model. That actually makes a lot of sense if you think of the estimation problem as the problem of first telling what distribution a given dataset was drawn from and then reading off the value of the estimand from that distribution. Let’s say that you have enough data to confidently rule out all but a small set of similar distributions. If the value of the estimand varies wildly over those distributions, you still can’t be very confident and should therefore report large uncertainty. Another way to understand the mechanism by which shrinking the model reduces uncertainty is that in smaller models sets of nearby distributions often have more similar values of the estimand.

3.6 Classical Theory

If you’ve taken a traditional statistics course you may have learned about Cramer-Rao and Fisher Information. In this section I’ll briefly explain how these topics are connected to efficiency theory for RAL estimators. If you’re not interested in this you should feel free to skip to the next chapter. None of this material is required to understand the rest of the book and I will try to be brief. More details can be found in Tsiatis (2007); I only bring it up to try and connect the prior knowledge some students have to the nonparametric context.

First we will apply all of the theory we’ve developed in this chapter to the parametric setting. After this is done we will make the promised connections to classical concepts.

3.6.1 Applying Nonparametric Results

A parametric model $\mathcal{P}$ is one in which every distribution can be put into one-to-one correspondence with an element $\theta \in \mathbb{R}^d$. To simplify the presentation we'll assume that every distribution has a density (with respect to Lebesgue measure). We write the density corresponding to the parameter value θ as p_θ .

In this context, the score function s is usually defined as the derivative of the log-likelihood with respect to θ : that is, a d -dimensional function of z . This is a natural extension of the score in a one-dimensional parametric model (i.e. a differentiable path), but the interpretation is a little different. For a one-dimensional model the score neatly encodes the “direction” in $\mathcal{L}_2$ in which the root density is moving as the single parameter changes. For a d -dimensional model, each of the d *elements* of the score encodes the same for each of the d parameters, others held constant. I find it helpful to keep the “path” interpretation and think of these as separate scores $s_1, \dots, s_d$ along d paths baked into one vector s . The tangent space is the linear span of these d scores, i.e. $\dot{\mathcal{P}} = \{c^\top s : c \in \mathbb{R}^d\}$.

In the parametric world we are most often interested in one of the parameters $\psi = \theta_1$ and consider the rest of the parameters $\eta = [\theta_2, \dots, \theta_d]$ to be “nuisances”. Therefore we write $\Psi(\mathbb{P}_\theta) = \psi$. For example, we might be interested in estimating the mean of a normal distribution but agnostic to the value of the standard deviation. In this context we can partition the elements of the score vector into $s = [s_\psi, s_\eta]$. These scores are typically not orthogonal: two different parameters can change the density in similar ways.

There are two useful identities that arise from these definitions. Firstly, if we move in the distribution along a path with score s_ψ , the pathwise derivative must exactly equal to 1: a one-unit change in the parameter induces a one-unit change in the parameter! Therefore, for any gradient ϕ ,

$$1 = \dot{\Psi}(s_\psi) = \mathbb{P}s_\psi\phi.$$

Secondly, if we move in any direction which does *not* change the parameter of interest ψ , the pathwise derivative must equal 0. These directions are precisely any linear combination of the $d - 1$ scores s_η , i.e. $\{c^\top s_\eta : c \in \mathbb{R}^{d-1}\}$. This set is referred to as the *nuisance tangent space* $\dot{\mathcal{P}}_{\mathbb{P},\eta}$. The resulting identity is then

$$0 = \dot{\Psi}(s) = \mathbb{P}s\phi \quad \text{for all } s \in \dot{\mathcal{P}}_{\mathbb{P},\eta}$$

In other words, any gradient is in the orthogonal complement of the nuisance tangent space and has inner product equal to one with the estimand score. These two observations completely triangulate the canonical gradient, as we will see. If we project the estimand score onto $\dot{\mathcal{P}}_{\mathbb{P},\eta}^\perp$ the result will be in the intersection of $\dot{\mathcal{P}}_{\mathbb{P},\eta}^\perp$ and the full tangent space $\dot{\mathcal{P}}_{\mathbb{P}}$. That intersection is

Combining these observations, the geometric meaning of $\mathbb{P}s_\psi \nabla \Psi = 1$ is that the length of the projection of the parameter score s_ψ onto the orthogonal complement of the nuisance tangent space $\mathring{\mathcal{P}}_{\mathbb{P}, \eta}$ is $\frac{1}{\|\nabla \Psi\|}$. (see Section ??). That projection $s^* = \mathring{\mathcal{P}}_{\mathbb{P}, \eta}^\perp s_\psi$ is typically called the *efficient score*.

Since we don't already have score vector in the orthogonal complement of the nuisance tangent space, the way to mechanically calculate the projection of s_ψ onto this space is to first project it onto the nuisance tangent space and then take the residual. If $\eta \in \mathbb{R}$,

Since we don't already have score vector in the orthogonal complement of the nuisance tangent space, the way to mechanically calculate the projection of s_ψ onto this space is to first project it onto the nuisance tangent space and then take the residual. If $\eta \in \mathbb{R}$,

If η is multidimensional the process is the same but the estimand score must be residualized against each score $s_{\eta_1}, s_{\eta_2}, \dots, s_{\eta_{d-1}}$ in turn. The formula is obviously more complicated so I'll omit it: there is no conceptual difference. In fact this even works if η is infinite-dimensional (such models are called *semi-parametric*).

$$\nabla\Psi = \frac{s^*}{\|s^*\|^2},$$

and thus the minimum asymptotic variance in this model is

$$\|\nabla\Psi\|^2 = \|s^*\|^{-2} = \left[\mathbb{P}s_\psi^2 - \frac{(\mathbb{P}s_\psi s_\eta)^2}{\mathbb{P}s_\eta^2} \right]^{-1}.$$

One benefit of such (semi-)parametric models is that the canonical gradient can usually be explicitly derived as shown above. The identities in the semiparametric model give different kinds of geometric information that each narrow down location of the gradient set Φ until it can be exactly triangulated. From the nonparametric theory we know the gradients make a linear variety that is orthogonal to the tangent space (after shifting to the origin) but we don't know exactly where it intersects the tangent space. Considering the nuisance scores tells us what “direction” or “axis” the point of intersection lies in within the tangent space (the orthogonal complement to the nuisance tangent space). Considering the estimand score tells us exactly how far along the intersection is in that direction.

3.6.2 Connection to Likelihood Theory

At last we can connect all of this to classical ideas in likelihood theory. The *Fisher information* matrix is defined as

$$I_{\mathbb{P}} = \mathbb{P}ss^\top = \mathbb{P} \begin{bmatrix} s_\psi^2 & s_\psi s_\eta^\top \\ s_\psi s_\eta & s_\eta s_\eta^\top \end{bmatrix}$$

which is nothing other than the variance-covariance matrix of the score vector s . I prefer to think of it as the set of norms and inner products of the d individual scores s_{θ_j} because that makes it clear that it is just a summary of some non-orthogonal basis of the tangent space: how long the different bases are and at what angles they are to each other. In classical theory the Fisher information is usually written with scores defined as derivatives of log-likelihoods, in which case the elements of the Fisher information also correspond to the expected value of the second or mixed derivatives of the log-likelihood under regularity conditions.

The Cramer-Rao theorem is a famous classical result that provides a lower bound for the asymptotic variance of estimators in parametric models. In a one-dimensional model the Cramer-Rao bound states that the asymptotic variance of any unbiased estimator $\hat{\psi}$ is greater than or equal to $1/I_{\mathbb{P}}$ ($I_{\mathbb{P}}$ is a scalar in this case). When $\psi = \theta_1$, as we have been discussing, the Cramer-Rao Theorem says that the asymptotic variance must be greater than or equal to the corner element $(I_{\mathbb{P}}^{-1})_{1,1}$ of the inverse Fisher information.

As before, we'll take $\eta \in \mathbb{R}$ to simplify the presentation. Applying the usual formula for inversion of a 2×2 matrix gives

$$(I_{\mathbb{P}}^{-1})_{1,1} = \frac{\mathbb{P}s_{\eta}^2}{\mathbb{P}s_{\psi}^2 \mathbb{P}s_{\eta}^2 - (\mathbb{P}s_{\psi}s_{\eta})^2} = \left[\mathbb{P}s_{\psi}^2 - \frac{(\mathbb{P}s_{\psi}s_{\eta})^2}{\mathbb{P}s_{\eta}^2} \right]^{-1}.$$

This is precisely the squared norm of the canonical gradient that we previously derived.

Another way to interpret the optimal asymptotic variance (squared norm of the canonical gradient) is as the supremum of the Cramer-Rao bounds over all differentiable paths. We've been considering estimands that are defined as a single component of the parameter vector θ but more generally we might consider estimands defined as arbitrary functions of the parameters, i.e. $\psi = f(\theta)$. In this context the Cramer-Rao bound at a distribution $\mathbb{P}_{\theta}$ is $\dot{f}_{\theta}^{\top} I_{\theta}^{-1} \dot{f}_{\theta}$ where $\dot{f}_{\theta} = \frac{d}{d\theta} \big|_{\tilde{\theta}=\theta} f(\tilde{\theta})$ and $I_{\theta} = I_{\mathbb{P}_{\theta}}$.

In the context of a differentiable path with parameter ϵ and score s , this simplifies to $\frac{\dot{\Psi}(s)^2}{\mathbb{P}s^2}$. Taking the suprema over differentiable paths (scores in the tangent space) gives exactly the square of the operator norm (see Definition ??) of the pathwise derivative, which is the squared norm of its the Riesz representer.

I personally find the parametric theory to be much more confusing than the nonparametric theory. To be fair, I have covered it very quickly and without the same level of pedagogical attention given to the other sections. Nonetheless, I do think there is something inherently more difficult about it. Having nuisance and target parameters and scores to keep track of complicates things, as does keeping track of regularity conditions needed for pointwise definitions of the scores. The nonparametric theory generalizes the traditional theory very nicely but in my opinion also makes it easier and more intuitive to understand!

4 Deriving Canonical Gradients

In the previous chapter we discovered that every pathwise-differentiable estimand has at least one RAL estimator and that any asymptotically optimal RAL estimator has an influence function equal to the canonical gradient (the *efficient* influence function; EIF). In the next part of the book we'll tackle the problem of constructing estimators that have this property, but, in order to do that we need to know what the canonical gradient is for any estimand and model of interest. In this chapter we'll therefore learn techniques to derive the canonical gradient.

There are many methods to derive canonical gradients that are more or less natural to depending on the kind of model and estimand. In this book we'll first focus on point mass contamination: a simple technique that works well in locally nonparametric models. In the process of illustrating this technique we'll cover some handy rules that make it easier to build canonical gradients up out of simpler blocks. Then we will discuss projection methods which are often useful when the model is restricted in some way and in particular for models where some factors of the joint distribution are known.

4.1 Point Mass Contamination

Point mass contamination (also referred to as the “Gateaux derivative” or “Hampel influence function” approach) is a easy-to-use strategy for deriving the gradient in nonparametric models (recall there is only one gradient in a nonparametric model). Nonparametric models are a good starting point because often we don't know anything useful about the data-generating distribution. Even if we do, nonparametric models are a reasonable fallback because we can still produce a good RAL estimator using the nonparametric EIF - it will just be less efficient than the efficient estimator under the restricted model that exploits our prior knowledge.

The idea of point mass contamination is to find paths in directions that make it easy to compute the value of the gradient at a given point z . We'll see that it suffices to take convex paths in the direction of distribution that is a point mass at a point z .

4.1.1 Discrete Distributions

Presume for a moment that we're working in a model where every variable is discrete but otherwise unspecified. When this is the case, the model is actually parametric because any